Measuring the Task Value of Precision in Closed-Loop Agents

Anonymous ACL submission

Abstract

A quantized agent reaches a decision and its BF16 reference is available. Should BF16 replace only this step, take over the rest of the episode, or both? End-to-end success cannot answer this. We measure the options directly: at a frozen state that replays bit-exactly, a 2×2 design crosses the source of the current action with the model that executes the rest of the episode, and a second W4 checkpoint of the same model serves as a same-precision reference substitution. Our central finding is that the ability of an intervention to change task outcomes and its ability to improve them have to be measured separately. On τ^2-bench with Qwen3-4B and Qwen3-8B, replacing a W4 agent's first diverging action with BF16's changes the outcome of 15.5–20.9% of task–block units, in both directions, for a net effect of -1.9 to $+2.9$ pp; in 26.9% of divergent states the swap's effect depends on which model continues. Swapping only the W4 calibration checkpoint, with no increase in bit width, also produces two-sided flips and continuation dependence of comparable size; the two substitutions share a core of rescued and harmed units on the action side, while their continuation-dependent flips are largely substitution-specific, and equivalence of the two turnovers within ± 3 pp is not shown (interim). A frozen uncertainty trigger finds outcome-changing decisions more often than random selection but captures little of the net gain a hindsight oracle shows is available. Evaluations of precision interventions for agents should therefore report sensitivity and value separately and include a same-precision reference substitution.

1 Introduction

A quantized LLM agent is cheaper to serve than its BF16 reference, and a deployment can keep the reference as a fallback, as dynamic precision routing does (Li et al., 2026c). Suppose the quantized agent (Q) reaches a decision at which the BF16 agent (B) would act differently. There are three ways to spend B here: **replace only this step, let B take over from the next step, or both**. Average task success does not answer this. On our checkpoints, quantized-minus-BF16 success ranges from -0.2 to $+4.2$ pp across four W4 cells, and an equivalence test at ± 3 pp fails in every cell (Appendix B); even a precise end-to-end number would not say which step an upgrade should touch.

Figure 1a shows one such decision from our data. In an airline task, Q looks up a reservation id that does not exist and receives "not found"; B looks up the user's profile. B's action looks better, and we can check what it is worth. We froze the state and ran all four combinations of {B's or Q's action} $\times$ {B or Q continuing}. Only B's action followed by B's continuation succeeds. When B supplies only this step, Q continues, searches for flights, writes a change and fails; when Q's failed lookup stands and B takes over, B fails too. The task value of an action is thus co-determined by the execution that follows it, and judging the action alone, however carefully, cannot reveal it.

We turn this observation into an evaluation. At a frozen state that replays bit-exactly, including the environment databases, the user simulator's view and the remaining step and error budget, we cross the source of the current action with the model that executes the rest of the episode (§3). The four branches give, on the same state, the value of upgrading this step, the rest of the episode, or both. To read these values, we add a same-precision reference substitution: the replacement action and continuation come from a second W4 checkpoint of the same model, calibrated on a different sample, instead of from B. We study Qwen3-4B and Qwen3-8B under simulated weight and KV-cache quantization in τ^2-bench (Barres et al., 2025), BFCL multi-turn and ALFWorld.

Sensitivity is not value. Two properties of an intervention are easily conflated. *Sensitivity* is its ability to change task outcomes: how many units flip (M) and how often the effect of the action depends on the continuation. *Value* is its ability to improve them: the net gain, which units it rescues and which it harms, and whether a selector can find the rescues in advance. Our central finding is that **the ability to change task outcomes and the ability to choose a beneficial precision intervention have to be measured separately.** The evidence has three links.

- **BF16 replacement is sensitive but has small net value** (§4). Replacing a W4 agent's first tool-structure-divergent action with B's changes the outcome in 15.5–20.9% of task×block units, with rescues and damages of similar size and a net effect of -1.9 to $+2.9$ pp (Figure 1b). In 26.9% of divergent states the swap has an effect under one continuation and none, or the opposite, under the other.

- **A same-precision substitution is about as sensitive, partly on different units** (§5). Swapping only the W4 calibration checkpoint also produces two-sided flips and continuation dependence of comparable size, so flips alone do not diagnose a need for precision. On the action side the two substitutions share a core of rescued and harmed units (107 units flip under both, against 30 expected by chance), yet 138 of the 245 flipping units flip under only one of them, and continuation-dependent flips are largely substitution-specific. Equivalence of the two turnovers within ± 3 pp is not shown.

- **Detecting sensitivity is not selecting value** (§6). A frozen Q-only uncertainty trigger captures more outcome-changing decisions than random selection, but they are rescues and damages alike: it captures 2–5 of the 34–38 net units that a hindsight oracle with the same budget shows are available.

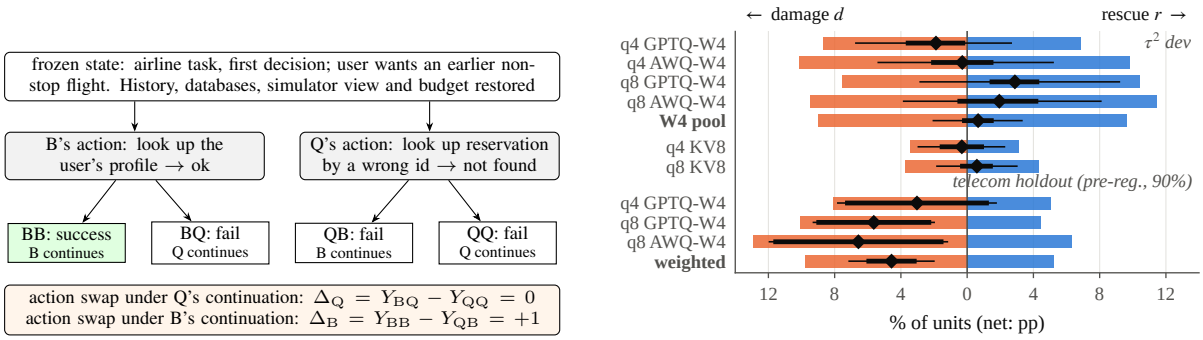
What should change in evaluation practice. A study that reports how often a precision change flips outcomes, or how strongly a model's uncertainty predicts flips, has measured sensitivity. To claim that a precision intervention is worth making, it must also report value, separately: the net gain with its rescue and harm counts and their denominators, under the execution that follows the intervention; and it must report the same quantities for a same-precision reference substitution, so that what higher precision adds can be told apart from what a change of model at the same precision already does.

Contributions. (1) *An evaluation methodology* that measures, on the same state, the task value of the current action, of the subsequent execution and of both, with a same-precision reference substitution for reading them. The factorial design is standard; the contribution is making the counterfactuals a precision decision needs executable and auditable in closed-loop agents (exact state restoration, budget accounting, replay acceptance, explicit denominators). (2) *Empirical findings* that separate sensitivity from value: an action's value varies with the execution that follows; a fixed-precision checkpoint swap produces sensitivity of comparable size; and a sensitivity signal locates outcome changes without yet converting them into net gain.

2 Related Work

Quantized agents compared episode by episode. *Flat Score, Amplified Failures* (Jang et al., 2026) is the closest study in setting: on τ^2-bench retail and telecom, 4-bit agents keep a flat task score while process damage grows, with a two-sided paired task split (33 tasks worse, 26 better for Gemma-4-31B on telecom) and recovery equally frequent at both precisions (55% after a failed call). Its episodes are sampled independently, so the precision of the action that went wrong and that of the controller that must recover change together; we hold the state fixed and vary the two separately. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

Forks and counterfactual replay. *The Replay Gap* (Gonuguntla, 2026) already forks live SWE-agent trajectories at a fixed step, continues them with a different model (Qwen3-4B FP8 vs. Qwen3-14B AWQ), and compares against same-model control forks; its 359 control forks had 0 success flips. Live forking and a same-model control are therefore not our novelty. In its design the new model produces the next action *and* the continuation, which is our "both" contrast; we intervene on the action and on the continuation separately. Restoring a state, replacing one action and letting



(a) (b)

Figure 1: Four-branch task value. (a) A real decision (Qwen3-4B, AWQ-W4; airline task 31, decision 1 of 12): B's action is worth a success only if B also continues; neither upgrading this step alone nor handing over the rest of the episode alone helps, so the task value of an action is co-determined by the execution that follows. (b) Replacing the first tool-structure-divergent action with B's, per cell: rescues r (right) and damages d (left) in % of all task×block units, and the net $G_{\text{all}} = r - d$ (diamond, pp). Thick/thin: fixed-pool/new-task 95% intervals on the τ^2-bench development panel; block-t/crossed 90% on the pre-registered telecom holdout ("weighted" is its decisional endpoint). All cells: Appendix Table 2.

the original policy continue locates decisive errors (Zhang et al., 2025), attributes failures through a do-intervention on one step (Shah, 2026), and turns alternative actions into training signal (Li et al., 2026a,b). This is our "this step only" contrast, and the factorial design itself is standard. We found no work among the 32 closest that crosses the action with the continuation at a precision-induced divergence or reads the result against a same-precision alternative.

Precision routing and its signals. DMR (Li et al., 2026c) learns a per-step BF16/W4 router from BF16-versus-quantized KL supervision and episode reward. Sensitivity has also been localised to phases (Lu et al., 2026) and task types (Zhang et al., 2026), and for quantized reasoning models to high-entropy, high-KL token positions (Lotfi et al., 2026). We do not propose a router. We measure the task value a router should learn, and our simple frozen trigger (§6) does not test, let alone refute, a learned router such as DMR.

Evaluation noise and user simulators. Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026) and with the user model (Seshadri et al., 2026; Jain and Sandhu, 2026).

3 Measuring the Task Value of a Replacement

3.1 Setting

Tasks and agents. We use τ^2-bench in text mode: a development panel of 58 retail and 27 airline tasks plus 80 telecom development tasks, and a holdout of 34 telecom tasks that no agent ran before its protocol was frozen. The agents are Qwen3-4B-Instruct (q4) and Qwen3-8B with thinking disabled (q8). One greedy Qwen3-30B-A3B-Instruct user simulator serves all agents. Retail and airline success is the final-database match; telecom uses the official task-basis success. BFCL multi-turn and ALFWorld, which have no user simulator, run once per item (Appendix A).

Quantization and serving. Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B: GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) at W4 and a KV8 cache in the main cells; GPTQ-W3, RTN-W3 and a rotated KV4 cache as boundary (stress) cells. B and Q run on vLLM with batch-invariant kernels and deterministic compilation in separate server processes, so no prefix cache is shared. A *block* is one closed-loop run of Q over the task pool under one prompt nonce (a random session-identifier line in the system prompt); blocks are the replicate unit.

3.2 Four branches on one state

Frozen state. Before Q's decision t we restore the state s_t: the full message history; the environ-

ment and user-device databases, rebuilt by strict replay of every recorded tool call; the user simulator's visible conversation; and the used step and error budget. B and Q generate their actions from identical prompt token ids, and the state is written to disk before any branch runs.

Crossed intervention. $Y_{\mathrm{XY}} \in \{0, 1\}$ is task success when the action at s_t comes from X and model Y executes the rest of the episode, X, Y $\in \{\mathrm{B}, \mathrm{Q}\}$ (Figure 1a). The action is executed by the real environment or answered by the real simulator, and all four branches start with the same remaining budget. Each upgrade option is a paired contrast with QQ:

$$G = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}} \quad \text{(this step only)},$$
$$H = Y_{\mathrm{QB}} - Y_{\mathrm{QQ}} \quad \text{(from the next step)},$$
$$F = Y_{\mathrm{BB}} - Y_{\mathrm{QQ}} \quad \text{(both)}.$$

The paired effect of the action swap under each continuation is $\Delta_{\mathrm{Q}} = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$ $(= G)$ and $\Delta_{\mathrm{B}} = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$; their difference $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}}$ is the action–continuation interaction. Each takes values in $\{-1, 0, +1\}$. When $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$ ("off-diagonal"), the swap's value depends on the continuation; a *strict reversal* is $\Delta_{\mathrm{Q}} = -\Delta_{\mathrm{B}} \neq 0$. For the action swap, a *rescue* is $(Y_{\mathrm{BQ}}, Y_{\mathrm{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$; with rates r and d, the turnover is $M = r + d$ and the net gain $r - d$. All contrasts measure effects on task success, not which action is correct.

Same-precision alternative. A large M or a frequent interaction could mean that this decision needs higher precision, or only that it is sensitive to any change of model. To tell these apart, we rerun the design with Q′, a second GPTQ-W4 checkpoint of the same model calibrated on a different sample, in place of B: the same Q episodes, tasks and nonce blocks, and a protocol frozen before any outcome (q4 and q8 GPTQ-W4). Turnover M', rescues r', damages d' and the off-diagonal share are defined as above with Q′ for B.

3.3 Making the counterfactuals auditable

Replay acceptance. A causal reading requires exact replay. Before any development outcome was analysed, an engineering gate on real divergence states passed every check: Q→Q no-ops matched in 20/20 states and B→B no-ops in 10/10, in every prompt-token hash, message, database hash, termination, reward and budget counter. In production, every fork task re-ran the QQ no-op on an outcome-independent 10% hash sample of its states, and one mismatch would have quarantined the task's whole output; all 924 sampled development states passed, as did all sampled states of the other banks and of the same-precision run (Appendix A). τ^2-bench resets step and error counters when it starts from a history; the fork restores them, so the remaining budget equals the original.

Which states. *Bank P*: for each task, block and cell, the first decision at which a_{B} and a_{Q} differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments; differences in text only are skipped. *Bank U*: one ordinary decision per task, drawn by a hash fixed before generation, in three nonce blocks, from Q's own trajectories (U1) or from trajectories shared by all cells of a model (U2B: BF16; U2Q: Q).

Denominators and inference. We never mix three denominators. (1) *Full pool*: every task×block unit, units without a Bank P state counting as zero; r, d, M and $G_{\mathrm{all}} = r - d$ use it. (2) *Divergent states*: the Bank P states of a cell; G, H, F and the off-diagonal share use it. (3) *Bank U*: one state per task. Within a cell all branches share each state and all tasks of a block share its nonce, so within-cell contrasts are paired and identified; cross-cell comparisons are descriptive, because each cell selects its own divergences. The primary interval treats the task pool as fixed (block bootstrap with at least 10 complete blocks, else t over block means); a crossed task×block bootstrap is the interval for new tasks. Only complete blocks enter.

4 Precision Replacement Redistributes Success and Failure

Many rescues, many damages, small net. Replacing an action with BF16's is a sensitive intervention of small net value. On the τ^2-bench development panel (85 tasks × 20 complete blocks), replacing a W4 agent's first tool-structure-divergent action with B's changes the task outcome in $M =$ 15.5–20.9% of units (Figure 1b). Rescues and damages are of similar size, $r =$ 6.8–11.4 and $d =$ 7.5–10.1 pp, so the net gain G_{all} lies between -1.9 and $+2.9$ pp. Two fixed-pool intervals exclude zero (negative for q4 GPTQ-W4, positive for q8 GPTQ-W4); no new-task interval does. Pooled over the four W4 cells, $G_{\mathrm{all}} = +0.7$ [-0.3,

+1.6] pp with $M = 18.6\%$. KV8 cells turn over fewer outcomes ($M = 6.5$–8.1%, a descriptive comparison). G_{all} is a local replacement effect, not an end-to-end comparison (Appendix B).

The same swap under two continuations. The net effect hides how the action's value depends on what follows. On divergent states, the action swap's average effect is the same under Q's and under B's continuation (W4 pool: $\Delta_{\text{Q}} = G = +0.5$ $[-0.7, +1.7]$, $\Delta_{\text{B}} = +0.5$ $[-1.0, +2.1]$ pp), and upgrading this step, the rest, or both has no detectable average gain ($H = +0.2$, $F = +0.7$ pp, all intervals covering zero). State by state, however, the two effects often differ. In 26.9% [25.7, 28.1] of pooled W4 divergent states (22.9–30.5% per W4 cell; new-task interval 22.1–32.0%), the swap has an effect under one continuation and a different one under the other; in the KV8 pool the share is 15.3% [13.2, 17.3]. Strict reversals are rare (3.2% of pooled W4 states); most off-diagonal states are an effect under one continuation and none under the other, as in Figure 1a. States where the swap helps more under Q ($I > 0$, 13.6%) and under B ($I < 0$, 13.4%) are about equally frequent, so the mean interaction is $+0.0$ $[-1.9, +2.0]$ pp although a quarter of states depend on the continuation. The 26.9% is one description of this coupling; what the design adds is that all four branches are measured on the same state, so the value of each upgrade option can be read off directly. Within single cells the options do differ: in q8 GPTQ-W4, $G - H = +2.9$ $[+1.3, +4.7]$ pp and $F - G = +2.1$ $[+0.2, +4.0]$; in q4 GPTQ-W4, B's action lowers success ($G = -2.9$ $[-5.7, -0.2]$, unadjusted). These orderings hold on the fixed task pool but not on new-task intervals, and no ordering transfers across cells (Appendix D).

Other states, tasks and environments. The pattern is not specific to τ^2-bench's simulated user or to development tasks. On BFCL multi-turn and ALFWorld, which have no simulated user (one run per item), the replacement turns over 5.2–24.6% of items in both directions, and the off-diagonal share is 4.7–17.1% in BFCL W4 cells and 32.8–40.0% in q8 ALFWorld W4 cells (post hoc). On 34 unseen telecom tasks, under a protocol frozen before any holdout unit ran (12 fresh blocks, three W4 cells, primary endpoint G_{all} on a crossed 90% interval against ± 3 pp), the replacement again rescued and damaged ($r = +5.21$ $[+3.30, +7.11]$, $d = +9.77$ $[+6.45, +13.10]$ pp) and task fragility repeated across independent nonce halves ($\rho = 0.59$, $p = .0001$). The primary estimate was $G_{\text{all}} = -4.57$ pp (crossed 90% $[-7.17, -1.96]$). The pre-registered readings had no negative category, so the decisional reading is *inconclusive*; outside them, the interval lies below zero. At ordinary (Bank U) states, turnover is lower (W4 cells, U1 and U2B: 4.7–9.0%), the net gain covers zero, and most outcome changes come from differences in text; their labels did not reproduce under a second user simulator, so we do not report them as a property of the action (Appendix H).

5 Sensitivity Without Higher Precision

A large turnover or a frequent interaction at a state is tempting evidence that the state needs higher precision. The same-precision reference substitution (§3.2) tests that reading: if these consequences come from precision, swapping one W4 checkpoint for another should produce markedly fewer of them. It also asks the value question on the same units: do the two substitutions rescue and harm the same episodes?

A higher bit width is not necessary for two-sided flips and continuation dependence. They also appear when only the W4 calibration checkpoint is swapped (Figure 2a). On the first 6 complete blocks (interim; the final analysis uses the longest complete block prefix at the evidence freeze), Q′ diverges from Q as often as B does (76.9% of units in both designs). Its action rescues $r' = 9.3$ [7.0, 11.7] and damages $d' = 8.4$ [6.4, 10.5] pp of units ($M' = 17.7\%$, against $M = 16.8\%$ for B), and 26.2% [20.7, 31.3] of its divergent states are off the diagonal, against 25.0% for B. Every one of these same-precision rates has a 95% lower bound above zero. The difference in turnover is $\Delta M = M - M' = -0.98$ pp (90% CI $[-3.26, +1.30]$). No precision-specific excess was detected, but equivalence within ± 3 pp is not shown either (two one-sided tests, $p = 0.067$), and the same holds for the net gain ($\Delta G_{\text{all}} = -1.57$ $[-5.17, +2.04]$, $p = 0.230$) and, descriptively, for the off-diagonal share (-1.21 $[-4.09, +1.67]$, $p = 0.133$; Figure 2b). We therefore do not say that the two designs are equivalent or that any change of model has the same effect. We say that flip rate and interaction size alone cannot diagnose a need for an upgrade: a fixed-precision swap already produces them at comparable size.

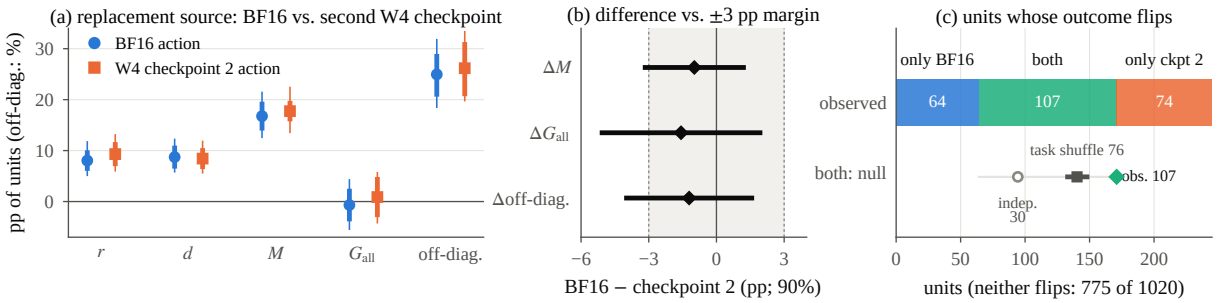


Figure 2: Same-precision reference (q4 and q8 GPTQ-W4 pooled; the same 1020 task×block units; interim, $K = 6$ blocks). (a) Replacing Q's first divergent action with B's (circles) or with a second W4 checkpoint's (squares): rescues r, damages d, turnover M and net G_{all} in pp of units, and the off-diagonal share in % of divergent states. Thick: fixed-pool 95% (t over block means); thin: new-task 95% (crossed). (b) Paired differences, B minus second checkpoint, with 90% intervals against the ±3 pp equivalence margin (shaded); no interval lies inside it. The off-diagonal difference is descriptive (different state sets). (c) Units whose outcome flips ($M = 1$ or $M' = 1$) under only B's action, both, or only the second checkpoint's; the lower row compares the observed both-flip count with its expectation if the two flip sets were independent within cell×block and with a within-task block shuffle that keeps each task's flip propensity (mean and 95% range).

Similar flip volume, overlapping but not identical units. Similar average turnover could come from the same or from disjoint sensitive episodes. It is neither (Figure 2c). Of 1020 units, 107 flip under both substitutions, against 30 expected if the two flip sets were independent within cell and block (Cohen's $\kappa = 0.53$ [0.46, 0.58]), and per-task flip rates correlate at $r = 0.68$ across the two designs. Sharing goes beyond task difficulty: shuffling the second design's flips across blocks within each task, which keeps every task's flip propensity, gives 76 [69, 84] both-flip units, against 107 observed. Yet 138 of the 245 flipping units flip under only one substitution. These overlap analyses are descriptive and were added after the interim result was seen.

A shared action-side core, substitution-specific continuation effects. Splitting each contrast into rescues and harms separates two regimes (Appendix Figure 7). On the action side (G), the two substitutions share a large core: 56 of B's 82 rescues and 51 of its 89 harms are also rescues and harms of Q′ ($\kappa = 0.60$ and 0.54), above a within-task shuffle (37 and 25 shared). Still, 56% of the units that flip under either substitution flip under only one, against 72% under the shuffle. On the continuation side (H) and under full substitution (F), overlap is at or near the shuffle level (H rescues $\kappa = 0.20$; F rescues and harms $\kappa = 0.34$ and 0.35, at chance under a stricter shuffle within task and QQ outcome). Continuation dependence is largely substitution-specific: among the 697 units that diverge in both designs, agreement on being off the diagonal is $\kappa = 0.29$ against 0.23 under the shuffle, and 70% of off-diagonal units are off-diagonal under one design only. Flip volume is similar for G and H; under full substitution B flips more units than Q′ (193 vs. 149; +4.3 [+0.4, +8.3] pp, one of several contrasts on six blocks). The net difference comes only from the substitution-specific units and is not resolved ($\Delta G = -1.6$ [−6.2, +3.0] pp). No unit is rescued under one design and harmed under the other, but this is forced: both designs share the Q episode, so it is no evidence. Sensitivity of the same size thus comes with a shared action-side core and largely substitution-specific continuation effects; only value measures on the same units reveal this.

The measured value of the BF16 replacement stands. The reference changes how B's consequences are read, not what they are. On the full 20 blocks, B's action is worth $G_{\text{all}} = +2.9$ pp in q8 GPTQ-W4 and -1.9 pp in q4 GPTQ-W4 on the fixed task pool (§4); on the interim same-precision blocks, B's and Q′'s net gains are -0.7 [−3.9, +2.5] and +0.9 [−3.1, +4.8] pp. A mean net gain near zero with an interval that covers zero is not a conclusion about value; whether B's action carries value beyond being a different plausible action is the question of the next comparison.

Common-state action sources. Above, each design selects its own divergent states. A preregistered comparison removes that difference: on 255 ordinary q8 states (85 tasks × 3 blocks, sam-

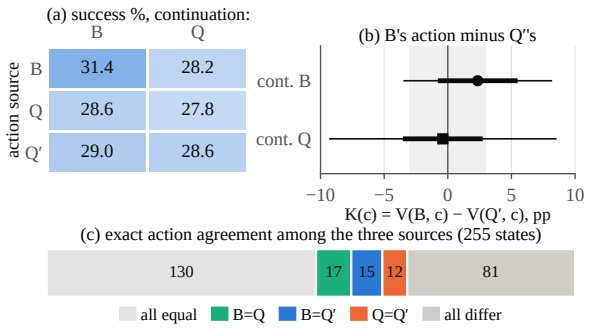


Figure 3: Common-state action sources (q8; 255 preregistered ordinary states, 3 blocks). (a) Success (%) by action source and continuation. (b) $K(c)$; thick: task-cluster bootstrap 95%, thin: t (block means) 95%; shaded: ± 3 pp. (c) Exact agreement of the three actions.

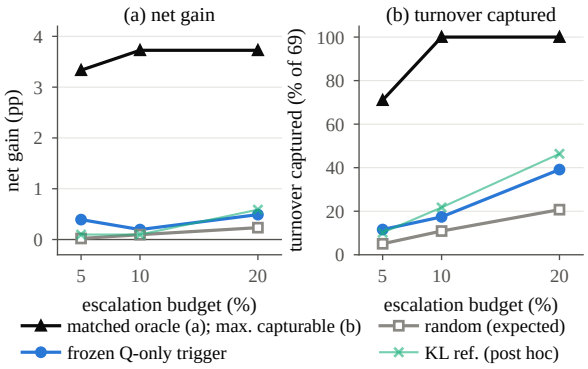


Figure 4: Sensitivity versus benefit, by escalation budget, on held-out ordinary states (1020 U1 rows; 38 rescues, 31 damages): a frozen Q-only uncertainty trigger, matched-count random selection (expected), a hindsight oracle with the same per-cell counts, and a faint post-hoc KL(B‖Q) reference that needs BF16. (a) Net gain, (rescues − damages)/1020, in pp; the oracle's gain is positive by construction. (b) Share of the 69 outcome-changing states selected; the black series is the most any selection with the same per-cell counts can capture. Values: Appendix Table 12.

pled before any outcome; none excluded), the current action comes from B, from Q (GPTQ-W4, first checkpoint) or from Q′ (second checkpoint), and B or Q continues. $K(c) = V(\mathrm{B}, c) - V(\mathrm{Q}', c)$ asks what B's action adds beyond another W4 action. A frozen rule claims value only if both 95% intervals exclude zero (Holm), equivalence only if both 90% intervals lie within ± 3 pp. B's action is not shown to be worth more than Q′'s (Figure 3b): under Q's continuation $K = -0.4$ pp (95%: t $[-9.3, +8.5]$, bootstrap $[-3.5, +2.7]$) and under B's $K = +2.4$ pp ($[-3.5, +8.2]$, $[-0.8, +5.5]$). Both are "not resolved": neither a difference nor equivalence is shown. Each contrast turns over 5.5–7.5% of states, and a second W4 checkpoint's action departs from Q's as often as B's does (agreement with Q's action 55.7% against 57.6%; B with Q′ 56.9%; Figure 3c). Any edge of B appears only when B also continues (31.4% against 29.0%; Figure 3a), consistent with continuation dependence (§4) but unresolved on three blocks (interaction $+2.7$ pp, 90%: t $[+0.5, +5.0]$, bootstrap $[-0.4, +5.9]$; secondary). Changing outcomes is thus not improving them, and without a same-precision reference B's turnover would read as a precision effect; this does not make precision irrelevant or the two actions equivalent.

6 Does Sensitivity Convert into Upgrade Value?

G, H and F require running B and all four branches; a deployable rule sees only Q. If outcome-changing decisions were the ones worth upgrading, a signal that finds them would suffice. We test whether it does.

Uncertainty locates divergence and outcome changes. Scored at the token level on one development block, Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically (AUC 0.66–0.91 per sub-bank and model), but among states where the actions differ, no Q-visible statistic reliably separates flips from non-flips (Appendix D.5). We then froze a Q-only trigger on that block before any later state was scored: escalate one decision when the minimum Q margin over its action is low, with per-cell thresholds at 5, 10 and 20% escalation rates. On two held-out blocks (1020 rows, 69 outcome-changing states) it captured 8, 12 and 27 outcome-changing states, against 3.48, 7.52 and 14.30 expected at random ($p = .0182$, .0613, .0002; Figure 4b).

It does not yet select net-beneficial replacements. The captured states were rescues and damages alike (6/2, 7/5 and 16/11), for net gains of $+4$, $+2$ and $+5$ against random expectations of $+0.21$, $+0.98$ and $+2.38$ (exact one-sided $p =$ .0345, .417, .262); no level was fixed in advance, and the 5% rate does not survive Holm correction. A hindsight oracle with the same per-cell counts nets $+3.33$, $+3.73$ and $+3.73$ pp (Figure 4a). This headroom is positive by construction, because the oracle picks only rescues and never a damage; the trigger captures 4/34, 2/38 and 5/38 of it. A post-

hoc KL(B‖Q) score, which needs BF16, also captures turnover above chance without a significant net gain (Holm $p = 1.0$). The held-out blocks reuse the development tasks. Read with §5, the result is consistent: a signal of where the model's action is fragile finds outcome changes, but outcome changes are two-sided and arise under a fixed-precision swap too, so finding them is not the same as finding where an upgrade pays. A learned router trained on task reward (Li et al., 2026c) is not tested here.

7 Where the Separation Holds or Changes

Each further configuration asks whether sensitivity and value separate as they do in the main cells; full matrices are in the appendix. Without a simulated user (BFCL, ALFWorld) and on unseen telecom tasks, the separation holds: turnover is two-sided and the net value is small or negative (§4).

A larger model: value appears. On BFCL, Qwen3-14B GPTQ-W4 gives $G = +5.7$ [+2.9, +8.7], $H = +0.4$ pp and 6.4% continuation-dependent states over 471 divergent items. Here the separation changes: replacing the action has a clearly positive net value, while continuation dependence is at the low end of the BFCL range. This is a single configuration within the Qwen3 family.

Lower bit widths: value moves to the continuation. We keep W3 and rotated KV4 separate from W4 as boundary checks. Four of the five stress cells turn over 18.4–20.4% of units with three blocks; in q8 RTN-W3, whose own success is low, B's continuation is what pays ($H = +15.3$, $F = +18.4$ against $G = -0.4$ pp). W2 runs did not complete and are not reported (Appendix G).

Local numerical quality versus task value. Rotation-based quantizers improve local numerics (Ashkboos et al., 2024; Liu et al., 2025). A matched W4A4 contrast of SpinQuant against QuaRot (same calibration, grid, clipping and KV cache) on shared q8 ordinary states asks whether that gain reaches action agreement and task value: [result pending: action effect with the BF16 continuation fixed, r, d, M and G_{all} with intervals, next to the KL difference with its interval]. We will not read a KL improvement without a task benefit as "decoupling" unless both effects are reported with sizes and intervals.

Stateful compression. KIVI (Liu et al., 2024) quantizes the KV cache that persists across turns, so a compressed state, not only an action, is carried forward. Applying the same four branches, with sensitivity and value reported separately: [result pending: KIVI-2 against KV8 on the shared states, with intervals].

A second family. Whether the action–continuation coupling and the separation of sensitivity from value appear outside Qwen3 is tested on Mistral-Small-3.2-24B once its fork gate passes: [result pending: off-diagonal share and r, d, M with intervals, or gate outcome]. An earlier Llama-3.1-8B check stopped at its gate because BF16 solved too few items to carry outcome variance (Limitations).

8 Discussion

Sensitivity and value are different measurements. Every analysis here points the same way. Replacing one action turns over many units in both directions for a small net effect, while the end-to-end differences of the four W4 cells can neither be distinguished from zero nor bounded within ±3 pp (§4). A second W4 checkpoint produces sensitivity of comparable size, partly on different units (§5), so a flip at a state is evidence that the state is fragile, not that it needs more bits. Interactions on deterministic states likewise arise without a change of bit width, and on identical states B's action is shown neither worth more than a second checkpoint's nor equivalent to it. A signal that finds flips does not yet find net gains (§6). A decision whose outcome changes easily when the model changes is therefore not thereby worth upgrading; the value of an upgrade must be judged against the complete replaced action, the execution that follows, and a same-precision alternative. In practice we recommend reporting M and continuation dependence as sensitivity, and the net gain with its rescue and harm counts as value, each with its denominator; keeping fixed-pool and new-task intervals apart; and reporting the same quantities for a same-precision reference substitution beside any claim about precision. We do not conclude that precision is irrelevant: the interim reference does not show the two substitutions to be equivalent, and B's measured net values in single cells stand.

Limitations

What we do not claim. We do not claim success equivalence between BF16 and W4 agents; equivalence tests at ± 3 pp on new-task intervals fail. G, H and F are oracle quantities, not the value of a deployable policy (§6). We do not claim that BF16 and a second W4 checkpoint are equivalent replacements. We do not claim a W$\times$KV interaction, and we do not present static precision restoration as a method (Appendix M).

One model family, two sizes. All crossed forks use Qwen3 models: 4B and 8B throughout, plus a single Qwen3-14B configuration on BFCL. Effects may differ at other scales and in other families. End-to-end runs also exist for Qwen3-14B and Qwen3-32B; apart from that one Qwen3-14B configuration they have no crossed forks, and we draw no conclusion from them (Appendix G). A time-boxed second-family check (Llama-3.1-8B-Instruct, GPTQ-W4, BFCL multi-turn) was stopped at its gate: BF16 solved 5 of 120 gate items, so the four arms carried almost no outcome variance, and it yields no second-family evidence.

Simulated quantization. Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

One user simulator; two small controls. All τ^2-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, cover 78 and 80 states, and say nothing about real users. The Bank P control has 20 states per cell, too few to confirm or refute few-pp contrasts. Neither control can be reproduced draw for draw. The replacement user is a hosted Sonnet 5.5 model sampled with its default settings, without temperature or seed control. In the Bank U control, Claude Code subagents answered many relay requests in one context, so a responder could see other arms or replicates of the same conversation. The Bank P control removed this exposure by issuing one isolated, single-request call per user turn. The same-precision reference (§5) covers q4 and q8 GPTQ-W4 on the development panel only, swaps one calibration sample for another and is reported on an interim block prefix; it is our only reference for how much turnover and continuation dependence a change of model without a change of bit width produces, and it says nothing about AWQ, KV8 or the stress cells.

LLM annotation. Process labels in Appendix K come from LLM annotators (Sonnet 5.5, two independent instances plus an adjudicator) and rule-based checks. No human annotation has been done. Agreement among instances of one model family shows reproducibility, not correctness.

Small confirmation and partial banks. The pre-registered holdout covers 34 telecom tasks and three cells. Its pre-registered readings had no negative category, so a clearly negative estimate could only be read as inconclusive; its negative estimate is established only for telecom, these three cells and this simulator. Most development results come from retail and airline, so the holdout confounds a domain shift (telecom) with generalisation to new tasks. On 80 telecom development tasks, the two q8 holdout cells show no detectable net effect (Appendix D.3), so the telecom domain alone does not produce a detectable drop; but their 6-block intervals are wide and q4 has no telecom development forks, so the holdout drop may reflect heterogeneity among its 34 tasks as much as a domain effect. Bank P intervenes only at the first tool-structure divergence per task and block. Bank U samples one state per task and block and has three blocks, so its block intervals rest on two degrees of freedom. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

Ethics Statement

This work involves no crowdworkers and no personal data. All users are simulated by an LLM on public benchmark tasks, and all agents are openly released Qwen3 checkpoints. The process annotations were produced by LLMs on synthetic benchmark conversations. The main risk we see is over-reading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

References

Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian L. Croci, Bo Li, Pashmina Cameron, Martin Jaggi, Dan Alistarh, Torsten Hoefler, and James Hensman. 2024. QuaRot: Outlier-free 4-bit inference in rotated LLMs. In *NeurIPS*.

Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. τ^2-Bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint arXiv:2506.07982*.

Bjarni Haukur Bjarnason, André Silva, and Martin Monperrus. 2026. On randomness in agentic evals. *arXiv preprint arXiv:2602.07150*.

Yihang Chen, Pin Qian, Su Wang, Chong Peng, Huan Xu, Xiyang Wu, and Yiqi Sun. 2026. Noise floor audit for agent benchmarks. *arXiv preprint arXiv:2608.22331*.

Peijie Dong, Zhenheng Tang, Xiang Liu, Lujun Li, Xiaowen Chu, and Bo Li. 2025. Can compressed LLMs truly act? an empirical evaluation of agentic capabilities in LLM compression. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Abhinav Dutta, Sanjeev Krishnan, Nipun Kwatra, and Ramachandran Ramjee. 2024. Accuracy is not all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. 2023. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *ICLR*.

Ashritha Gonuguntla. 2026. The replay gap: Static evaluation of model switching in LLM agents scores the wrong world. *arXiv preprint arXiv:2608.08239*.

Ashish Jain and Armaan Sandhu. 2026. UserProxyBench: Evaluating LLM user simulators for agent benchmarks and training. *arXiv preprint arXiv:2609.38043*.

Jiwon Jang, Kisu Yang, Heuiseok Lim, and Hyunwoo Park. 2026. Flat score, amplified failures: How the error budget masks damage in quantized LLM agents. *arXiv preprint arXiv:2607.27275*.

Mukai Li, Qingcheng Zeng, Tianqing Fang, Zhenwen Liang, Linfeng Song, Qi Liu, Haitao Mi, and Dong Yu. 2026a. Verified critical step optimization for LLM agents. *arXiv preprint arXiv:2602.03412*.

Yuanhao Li, Hongbo Wang, Xuhong Chen, Yiming Cao, and Xunzhu Tang. 2026b. Counterfactual rollout replay: Forkable environments as free process rewards for software engineering agents. *arXiv preprint arXiv:2609.33875*.

Yuanzhe Li, Jianing Deng, Jingtong Hu, Tianlong Chen, Song Wang, and Huanrui Yang. 2026c. Dynamic mixed-precision routing for efficient multi-step LLM interaction. *Preprint*, arXiv:2602.02711.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. 2024. AWQ: Activation-aware weight quantization for llm compression and acceleration. In *MLSys*.

Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. 2025. SpinQuant: LLM quantization with learned rotations. In *ICLR*.

Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. 2024. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. In *ICML*.

Sanae Lotfi, Polina Kirichenko, Steven Li, and Zechun Liu. 2026. Quantized reasoning models think they need to think longer, but they do not. *arXiv preprint arXiv:2606.00206*.

Haiquan Lu, Zigeng Chen, Gongfan Fang, Xinyin Ma, and Xinchao Wang. 2026. Mix-Quant: quantized prefilling, precise decoding for agentic LLMs. *arXiv preprint arXiv:2605.20315*.

Aditi Patodiya. 2026. Same request, different answer: Quantization amplifies Cache-Induced divergence in LLM serving. *arXiv preprint arXiv:2609.04748*.

Preethi Seshadri, Samuel Cahyawijaya, Ayomide Odumakinde, Sameer Singh, and Seraphina Goldfarb-Tarrant. 2026. Lost in simulation: LLM-simulated users are unreliable proxies for human users in agentic evaluations. *arXiv preprint arXiv:2601.17087*.

Jaineet Shah. 2026. Causal agent replay: Counterfactual attribution for LLM-agent failures. *arXiv preprint arXiv:2606.08275*.

Guibin Zhang, Junhao Wang, Junjie Chen, Wangchunshu Zhou, Kun Wang, and Shuicheng Yan. 2025. AgenTracer: Who is inducing failure in the LLM agentic systems? *arXiv preprint arXiv:2509.03312*.

Manyi Zhang, Ji-Fu Li, Zhongao Sun, Xiaohao Liu, Zhenhua Dong, Xianzhi Yu, Haoli Bai, and Xiaobo Xia. 2026. QuantClaw: precision where it matters for OpenClaw. *arXiv preprint arXiv:2604.22577*.

A Setup Details

Splits. The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from τ^2-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

Success endpoints. For retail and airline we use the final-database match with normal termination. This is not τ^2-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures. Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

Fork mechanics. The forced action is returned by a fork agent for the one step that follows h_t. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. τ^2-bench resets step and error counters when it starts from a history. The fork restores them to the values used at t, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount.

Replay acceptance details. In the engineering gate, one B→B field, the official reward, was missing from older trajectories and was recomputed from the stored reward information before the verdict. Besides the 924 sampled development states, 328/328 sampled Bank U states passed the QQ no-op, and the Bank U gate passed a further 135/135 no-ops. Serving in separate processes closes the prefix-cache channel (Patodiya, 2026). The same-precision run's acceptance counts are in Appendix D.6.

User simulator. The served user simulator is Qwen3-30B-A3B-Instruct in every episode; a user-model label stored in the fork state files is outdated and should be ignored.

B End-to-End Success Comparison

The replacement gains G_{all}, G, H and F are not end-to-end comparisons. Table 1 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

C Full Tables: Rescue, Damage and the Four Branches

Table 2 gives every cell of Figure 1, including the stress cells, with new-task intervals. Table 3 gives every cell of Figure 5, with $F-G$, $F-H$ and the $(\Delta_{\text{Q}}, \Delta_{\text{B}})$ summaries.

Cell	Q / BF16 (%)	Q−BF16 [95% CI]	NI 3 pp	TOST ±3 pp
q4 GPTQ-W4	40.0 / 35.8	+4.2 [−2.4, +10.5]	pass	fail
q4 AWQ-W4	36.8 / 35.8	+0.9 [−5.8, +7.5]	fail	fail
q8 GPTQ-W4	30.1 / 30.4	−0.2 [−7.7, +6.8]	fail	fail
q8 AWQ-W4	31.5 / 30.4	+1.2 [−5.3, +7.6]	fail	fail
pool, all four		+1.5 [−2.7, +5.6]	pass	fail

Table 1: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence.

Cell	units	div.	r	d	M	G_{all} [fixed pool]	[new tasks]
q4 GPTQ-W4	85×20	1102	6.8	8.7	15.5	−1.9 [−3.7, −0.1]	[−6.8, +2.7]
q4 AWQ-W4	85×20	1439	9.8	10.1	19.9	−0.3 [−2.2, +1.6]	[−5.4, +5.2]
q8 GPTQ-W4	85×20	1473	10.4	7.5	17.9	+2.9 [+1.4, +4.4]	[−2.9, +9.2]
q8 AWQ-W4	85×20	1562	11.4	9.5	20.9	+1.9 [−0.6, +4.3]	[−3.9, +8.1]
W4 pool	85×20		9.6	9.0	18.6	+0.7 [−0.3, +1.6]	
q4 KV8	85×22	593	3.1	3.4	6.5	−0.3 [−1.7, +1.0]	[−3.0, +2.3]
q8 KV8	85×22	855	4.3	3.7	8.1	+0.6 [−0.4, +1.6]	[−1.9, +3.0]
q4 GPTQ-W3^s	85×3	232	9.0	10.2	19.2	−1.2 [−7.0, +4.7]	[−8.6, +5.9]
q4 KV4-rot^s	85×3	233	11.8	7.1	18.8	+4.7 [−4.1, +13.5]	[−3.1, +12.9]
q8 GPTQ-W3^s	85×3	242	10.2	8.2	18.4	+2.0 [−12.5, +16.4]	[−6.3, +11.8]
q8 KV4-rot^s	85×3	239	11.8	8.6	20.4	+3.1 [−14.5, +20.8]	[−5.9, +12.9]
q8 RTN-W3^s	85×3	255	3.1	3.5	6.7	−0.4 [−3.8, +3.0]	[−4.7, +3.9]

Table 2: Replacing Q's first tool-structure-divergent action while Q continues, τ^2-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix B). Fixed pool: block bootstrap (≥10 blocks) or t over block means; new tasks: crossed task×block bootstrap. ^{s}stress cell, 3 blocks.

D Supplementary Results for the Main Findings

This appendix gives in full the analyses that the main text summarises.

D.1 Upgrade options within cells

On average over W4 cells, no option has a detectable gain. Pooled over the four W4 cells (Figure 5), $G = +0.5$, $H = +0.2$ and $F = +0.7$ pp, and every interval covers zero, as do their differences.

Within cells, the paired gains differ. In q8 GPTQ-W4, $G = +3.3$ [+1.6, +5.0] pp exceeds $H = +0.4$ ($G - H = +2.9$ [+1.3, +4.7]), and $F = +5.4$ exceeds G ($F - G = +2.1$ [+0.2, +4.0]). In q8 AWQ-W4, $G - H = +3.4$ [+1.5, +5.2] and $F - G = -1.5$ [−4.2, +0.8]. In q4 GPTQ-W4, $G = -2.9$ and $G - H = -3.6$ [−6.6, −0.6]. In the stress cell q8 RTN-W3 (three blocks), $H = +15.3$ and $F = +18.4$ against

Cell	states	G (this step)	H (next step on)	F (from here on)	$G - H$	$F - G$	$F - H$ ($= \Delta_B$)	off-diag. %	rev. %	mean I
q4 GPTQ-W4	1102	−2.9 [−5.7, −0.2]	+0.7 [−1.6, +3.0]	−2.5 [−5.3, +0.2]	−3.6 [−6.6, −0.6]	+0.5 [−1.2, +2.2]	−3.2 [−6.4, +0.2]	25.0	2.6	+0.3
q4 AWQ-W4	1439	−0.3 [−2.6, +1.9]	+0.9 [−1.1, +2.9]	−0.8 [−3.3, +1.6]	−1.3 [−3.0, +0.6]	−0.4 [−2.4, +1.5]	−1.7 [−3.6, +0.4]	29.3	3.1	+1.3
q8 GPTQ-W4	1473	+3.3 [+1.6, +5.0]	+0.4 [−1.2, +1.9]	+5.4 [+3.5, +7.2]	+2.9 [+1.3, +4.7]	+2.1 [+0.2, +4.0]	+5.0 [+3.2, +6.8]	22.9	2.7	−1.7
q8 AWQ-W4	1562	+2.1 [−0.6, +4.7]	−1.3 [−3.5, +0.9]	+0.6 [−1.5, +2.7]	+3.4 [+1.5, +5.2]	−1.5 [−4.2, +0.8]	+1.9 [−0.1, +3.8]	30.5	4.2	+0.3
W4 pool	5576	+0.5 [−0.7, +1.7]	+0.2 [−1.3, +1.6]	+0.7 [−0.6, +2.1]	+0.4 [−0.8, +1.7]	+0.2 [−1.0, +1.2]	+0.5 [−1.0, +2.1]	26.9	3.2	+0.0
q4 KV8	593	−1.0 [−5.3, +3.2]	−2.5 [−4.5, −0.5]	−1.2 [−4.7, +2.3]	+1.5 [−3.1, +6.0]	−0.2 [−3.5, +3.7]	+1.3 [−2.5, +4.9]	15.7	1.2	−2.4
q8 KV8	855	+1.3 [−0.9, +3.4]	−0.4 [−2.1, +1.5]	+2.5 [+0.0, +5.1]	+1.6 [−0.6, +3.9]	+1.2 [−0.5, +3.2]	+2.8 [+0.3, +5.5]	14.9	1.2	−1.5
KV8 pool	1448	+0.1 [−2.8, +3.0]	−1.4 [−2.7, −0.1]	+0.6 [−1.8, +3.1]	+1.6 [−1.2, +4.1]	+0.5 [−1.6, +2.8]	+2.1 [−0.2, +4.2]	15.3	1.2	−1.9
q4 GPTQ-W3^s	232	−1.3 [−7.7, +5.0]	+4.3 [−10.7, +19.4]	+1.3 [−1.9, +4.4]	−5.6 [−15.0, +3.7]	+2.6 [−0.6, +5.8]	−3.0 [−15.1, +9.1]	31.5	3.9	+1.7
q4 KV4-rot^s	233	+5.2 [−4.5, +14.9]	+9.4 [+5.1, +13.7]	+12.4 [+11.0, +13.9]	−4.3 [−18.3, +9.8]	+7.3 [−2.0, +16.5]	+3.0 [−1.9, +8.0]	33.0	1.7	+2.1
q8 GPTQ-W3^s	242	+2.1 [−13.3, +17.2]	+0.4 [−7.4, +8.2]	+2.1 [−7.4, +11.4]	+1.7 [−9.1, +12.3]	+0.0 [−11.0, +11.1]	+1.7 [−0.1, +3.4]	23.1	4.5	+0.4
q8 KV4-rot^s	239	+3.3 [−15.4, +21.9]	−0.8 [−18.2, +16.3]	+4.6 [−16.0, +25.0]	+4.2 [+2.5, +5.8]	+1.3 [−1.8, +4.3]	+5.4 [+0.8, +10.0]	31.0	4.6	−2.1
q8 RTN-W3^s	255	−0.4 [−3.8, +3.0]	+15.3 [+0.7, +29.9]	+18.4 [+16.7, +20.1]	−15.7 [−27.5, −3.9]	+18.8 [+13.8, +23.9]	+3.1 [−13.0, +19.2]	29.0	0.4	−3.5

Table 3: Three replacement gains and their paired differences on each cell's Bank P states (pp; same denominator in every column). $G = Y_{BQ} - Y_{QQ}$, $H = Y_{QB} - Y_{QQ}$, $F = Y_{BB} - Y_{QQ}$. Off-diag.: share of states with $\Delta_Q \neq \Delta_B$; rev.: strict reversals. Fixed-pool 95% intervals (block bootstrap over 20–22 complete blocks; t over 3 blocks for ^{s}stress cells, which cannot be pooled). New-task intervals are in Table 5. Within a row, contrasts are paired and identified; between rows, descriptive.

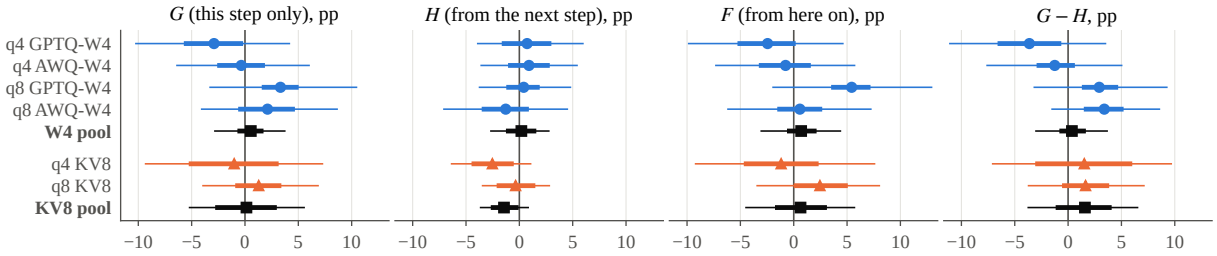


Figure 5: Replacement gains G, H, F and $G - H$ per core cell (pp; each cell's Bank P divergent states). Thick: fixed-pool 95% (block bootstrap, 20–22 blocks); thin: new-task 95% (crossed task×block). Squares: equal-weight pools. Within a row, contrasts are paired; between rows, descriptive. Stress cells, $F - G$, $F - H$ and (Δ_Q, Δ_B) summaries: Table 3.

$G = -0.4$ ($G - H = -15.7$ [−27.5, −3.9]). In q4 KV8, B continuing after Q's action lowers success ($H = -2.5$ [−4.5, −0.5], unadjusted). After Holm adjustment over the four core W4 cells, $G - H$ stays below .05 in both q8 cells ($p = .013$, .012) but not in q4 GPTQ-W4 ($p = .069$). These within-cell orderings are exploratory: they hold on the fixed 85 tasks but are not confirmed on new tasks (Table 5) or under a second simulator (Appendix D.4), and cells select different states.

A BF16 reference action can lower success. In q4 GPTQ-W4, B's action lowers success under Q's continuation on the fixed task pool ($G = -2.9$ [−5.7, −0.2] pp, unadjusted; the new-task interval covers zero), with a similar point estimate under B's ($\Delta_B = -3.2$ [−6.4, +0.2]).

D.2 Continuation dependence per cell and environment

Figure 6 gives the off-diagonal share per cell; the pooled (Δ_Q, Δ_B) tables are in Figure 8. Because replay is exact, each off-diagonal state shows a real interaction on the success scale, not replay noise. Cohen's κ between Δ_Q and Δ_B, against a marginal-preserving re-pairing of states, is 0.29 in the W4 pool; it measures how well the two effects agree across states, not whether the interaction is significant (Appendix F.1). Strict sign reversals are 1.2–4.2% of states in each W4 and KV8 cell, or 8–14% of the off-diagonal mass. In BFCL multi-turn and ALFWorld (post hoc; one run per item; Table 7), the off-diagonal share is 4.7–17.1% in the four BFCL W4 cells (pool 10.3%) and 32.8–40.0% in the two q8 ALFWorld W4 cells (pool 36.4%). Continuation dependence therefore does not require an LLM user simulator, and its size differs by environment. The net replacement gain on these benchmarks covers zero except for BFCL q4 AWQ-W4, where it is −3.8 [−6.5, −1.3] pp. The value of a fixed replacement policy is $E[G]$, $E[H]$ or $E[F]$, not $E[I]$. All 16 outcome patterns are in Appendix E.

D.3 Telecom: development tasks and the holdout's four branches

The holdout protocol, its exclusions and strata are in Appendix L; all 12 blocks were complete, and 1200 of the 1224 units were analysed after 2 structural task×cell exclusions. The cells were q4 GPTQ-W4, q8 GPTQ-W4 and q8 AWQ-W4,

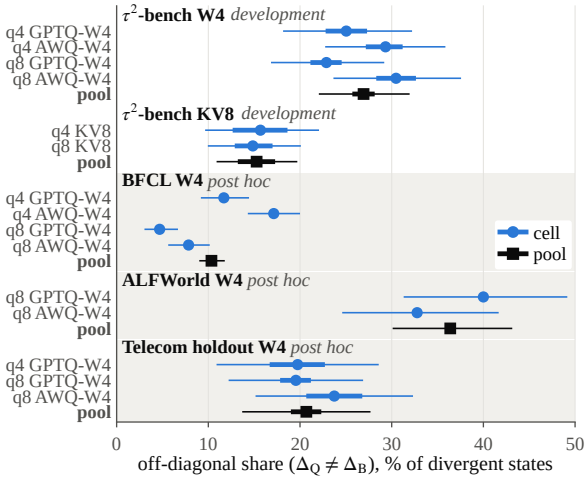


Figure 6: Same action, different value: per-cell share of divergent states with $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$. Thick: fixed-pool 95% (development: block bootstrap; holdout: block t); thin: new-task 95% (crossed; BFCL/ALFWorld: item bootstrap). Pools: equal weight (holdout: pre-registered weights). Shaded groups are post hoc. Cross-cell comparisons are descriptive.

weighted 1/2, 1/4, 1/4; the block-t 90% interval of the primary endpoint is $[-6.08, -3.05]$. On 80 telecom development tasks (6 blocks; fixed-pool 95%), the two q8 holdout cells give $G_{\mathrm{all}} = +2.7\ [-4.7, +10.1]$ (GPTQ-W4) and $-0.6\ [-4.9, +3.7]$ pp (AWQ-W4), against -5.6 and -6.6 on the holdout; q4 has no telecom development forks. A post hoc four-branch analysis of the 1180 holdout divergent states (Table 18) gives $G = -4.6\ [-7.8, -1.5]$, $H = -2.8\ [-6.3, +0.6]$, $F = -2.5\ [-6.2, +1.3]$, $G - H = -1.8\ [-4.8, +1.1]$ and mean $I = -5.0\ [-9.6, -0.4]$ pp (crossed 95%), unlike the near-zero development W4 pool, with 20.7% of states off the diagonal (Figure 6). This G conditions on divergence and nearly equals G_{all} because 1180 of 1200 units diverge. In an exploratory analysis of the development data, the top 20% of tasks hold 57–71% of all flips in the W4 and KV8 cells (Appendix I).

D.4 Ordinary states and the user simulator

Ordinary states turn over less, mostly through text under our simulator. At ordinary states, replacing Q's action changes the outcome less often than at first tool-structure divergences (a descriptive comparison of different state populations, with no interval on the difference). Over blocks f01–f03 (task-equal; t intervals over block means), M per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within ± 2.8 pp (U1, U2B) and ± 5.0 pp (U2Q), and every interval covers zero (Appendix H). Identical actions never change the outcome, as determinism implies. After deduplication, 150 of the 208 outcome-changing cell×state rows (≈72.1%) are class (ii), the same tool structure with different text; 26 involve a different tool name, call count or argument (Table 10). The rows are not independent states: the three blocks repeat the same tasks, and each U2B/U2Q state appears once per cell of a model. This is a *composition share*: per instance, text, tool/argument and act/talk differences flip the outcome in 14.4%, 18.2% and 18.9% of rows. Text dominates because it is common (1045 of 1357 differing rows), not because each text difference matters more. Among simple content features, only an above-median length difference between the two texts co-occurs with outcome change (in all four U1/U2B × model groups); question, confirmation-request and new-entity flags show no consistent association. These are untested regular-expression features. Bank P cannot see these differences by design.

Sensitivity to the user simulator. Text actions act through the user simulator, and changing the simulator moves τ-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated class-(ii) flips of block f01 and 39 matched non-flip states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5, answering through a file relay from Claude Code subagents (reproducibility limits in the Limitations). Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** The control changes three things at once (simulator model, sampling, relay interface), so it cannot say which of them changed the labels.

Sensitivity of the four-branch results. Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms

five times each with Sonnet 5.5 as the user, one isolated single-request call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs (correlation 0.63–0.68). Pooled over cells, $G = -3.0$ $[-10.2, +4.8]$, $H = +5.5$ $[+0.0, +11.2]$ and $G - H = -8.5$ $[-14.8, -2.8]$ pp under Sonnet, against $G - H = -2.5$ $[-11.2, +6.2]$ on the same states under the original simulator (change -6.0 $[-16.2, +4.2]$). With 20 states per cell, this control cannot resolve the few-pp contrasts of Table 3 (on this sample even the original simulator gives $G - H = 0.0$ in both q8 W4 cells), so it neither confirms nor reverses the within-cell orderings. Per-cell estimates under both simulators are in Figure 9, and an exploratory mean squared interaction under Sonnet is in Appendix F.2.

Bank P and Bank U measure different quantities. Bank P skips text-only differences by design, yet these make up 150 of the 208 outcome-changing rows at ordinary states, and their labels did not reproduce under a second simulator (Appendix H; §4). Including text actions requires a simulator control before their consequences are reported as a property of the action.

D.5 Token-level analysis

Following Lotfi et al. (2026), we scored every Bank U f01 state at the token level (1140 cell × state rows, 58 flips), teacher-forcing both models on the generated ids of a_{B} and a_{Q}. We record truncated top-20 entropy, the top-1 margin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

Uncertainty is associated with divergence. Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified).

Among divergent states, it does not reliably identify consequences. Restricting to states where the actions differ removes the "any difference at all" signal. In U1 the signal is not detectable: no Q-visible statistic has an interval excluding 0.5 (21 and 9 flips; f01). Only small sub-banks keep a signal: q4 U2B (mean entropy over a_{Q}, AUC 0.73 [0.53, 0.89]) and q4 U2Q (entropy and margin at the action start, AUC 0.90 and 0.10; 10 states, 4 flips). At the first divergent token, pooled entropy and KL do not separate flips from non-flips except in q4 U2Q (KL AUC 0.80 [0.65, 0.93], entropy 0.25 [0.07, 0.48]; 4 flips). Per cell, two further unadjusted intervals exclude 0.5, in opposite directions (U1 q4 GPTQ-W4 entropy 0.30; U2B q4 KV8 KL 0.10). No class of first divergent token reproduces the marker enrichment of Lotfi et al. (2026). The largest q4 enrichment is ordinary word choice (rate ratio 1.89, Fisher $p = .019$, not corrected). On development data, uncertainty is associated with *where* quantization changes the action more strongly than with *which* changes matter. The frozen trigger of §6 uses held-out blocks f02/f03 (1020 U1 rows).

D.6 Same-precision reference per cell

Per model (interim, $K = 6$), the turnover difference is $\Delta M = +0.6$ $[-3.1, +4.3]$ pp in q4 and -2.5 $[-5.6, +0.5]$ in q8 (fixed pool, 95%), with crossed intervals $[-5.5, +6.9]$ and $[-7.3, +2.7]$. Within ± 3 pp, equivalence is shown for no per-model ΔM or ΔG_{all}; the q8 off-diagonal difference ($+0.40$, 90% $[-2.12, +2.92]$) is the only per-model contrast inside the margin. The protocol originally mapped "interval includes 0 and $|\Delta M| \leq 3$ pp" to "similar"; after the interim result had been seen, an amendment replaced that reading by two one-sided tests, which are stricter, and changed no estimate. Acceptance: every state in both designs passed Q replay and every branch passed the prefix and environment checks; QQ no-op samples passed 265/265 (precision) and 162/162 (same precision), and no directory was quarantined. At the unit level, κ is 0.43 in q4 and 0.61 in q8; at the task level, cross-design agreement at matched noise is 0.765 of within-design replication.

E All 16 Outcome Patterns and New-Task Intervals

Table 4 lists every four-branch pattern per cell; `0000` (39–58%) and 1111 (6–20%) dominate. Table 5 gives the new-task intervals for Table 3.

F Supplementary Four-Branch Analyses

F.1 Marginal-preserving pairing agreement control

For each cell, observed agreement is the diagonal share of the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ table, $p_o = \sum_k P(\Delta_{\mathrm{Q}} = k, \Delta_{\mathrm{B}} = k)$, so the off-diagonal share is $1 - p_o =$

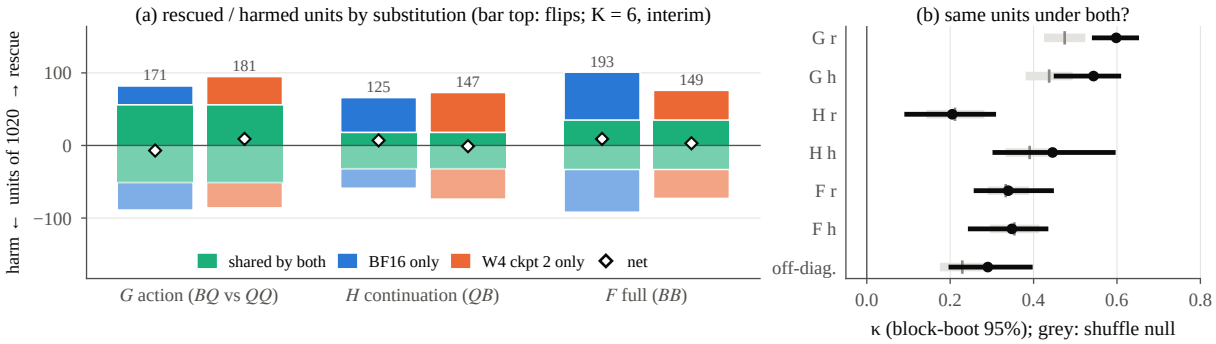


Figure 7: Which units the two substitutions rescue and harm (q4 and q8 GPTQ-W4 pooled; 1020 units; interim, $K = 6$; descriptive). (a) Per contrast (G: action side, BQ vs. QQ; H: continuation side, QB vs. QQ; F: full substitution, BB vs. QQ) and substitution (B, or the second W4 checkpoint in the B role): rescues above zero and harms below, split into units shared by both substitutions and substitution-specific units; the bar-top number is the flip count and the diamond the net. (b) Cohen's κ of each set between the two substitutions (block-bootstrap 95%) against a shuffle null (grey: within task and QQ outcome for rescue and harm sets, within task for the off-diagonal set). Rescue under one substitution and harm under the other cannot co-occur, because the QQ episode is shared.

$(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$	q4 AWQ	q4 GPTQ-W3	q4 GPTQ	q4 KV4r	q4 KV8	q8 AWQ	q8 GPTQ-W3	q8 GPTQ	q8 KV4r	q8 KV8	q8 RTN-W3
0000	634 (44.1%)	105 (45.3%)	472 (42.8%)	92 (39.5%)	286 (48.2%)	767 (49.1%)	130 (53.7%)	766 (52.0%)	113 (47.3%)	497 (58.1%)	141 (55.3%)
0001	55 (3.8%)	7 (3.0%)	29 (2.6%)	3 (1.3%)	14 (2.4%)	66 (4.2%)	2 (0.8%)	36 (2.4%)	8 (3.3%)	15 (1.8%)	3 (1.2%)
0010	57 (4.0%)	13 (5.6%)	54 (4.9%)	13 (5.6%)	11 (1.9%)	38 (2.4%)	8 (3.3%)	34 (2.3%)	10 (4.2%)	15 (1.8%)	20 (7.8%)
0011	58 (4.0%)	6 (2.6%)	71 (6.4%)	4 (1.7%)	37 (6.2%)	30 (1.9%)	9 (3.7%)	49 (3.3%)	3 (1.3%)	39 (4.6%)	0 (0.0%)
0100	50 (3.5%)	6 (2.6%)	36 (3.3%)	9 (3.9%)	8 (1.3%)	53 (3.4%)	8 (3.3%)	45 (3.1%)	13 (5.4%)	20 (2.3%)	1 (0.4%)
0101	21 (1.5%)	4 (1.7%)	17 (1.5%)	4 (1.7%)	11 (1.9%)	36 (2.3%)	7 (2.9%)	12 (0.8%)	5 (2.1%)	5 (0.6%)	2 (0.8%)
0110	22 (1.5%)	5 (2.2%)	14 (1.3%)	2 (0.9%)	2 (0.3%)	33 (2.1%)	4 (1.7%)	16 (1.1%)	2 (0.8%)	5 (0.6%)	1 (0.4%)
0111	41 (2.8%)	5 (2.2%)	21 (1.9%)	7 (3.0%)	13 (2.2%)	51 (3.3%)	6 (2.5%)	31 (2.1%)	7 (2.9%)	7 (0.8%)	6 (2.4%)
1000	51 (3.5%)	8 (3.4%)	32 (2.9%)	19 (8.2%)	16 (2.7%)	67 (4.3%)	10 (4.1%)	75 (5.1%)	16 (6.7%)	28 (3.3%)	32 (12.5%)
1001	23 (1.6%)	4 (1.7%)	15 (1.4%)	2 (0.9%)	5 (0.8%)	33 (2.1%)	7 (2.9%)	24 (1.6%)	9 (3.8%)	5 (0.6%)	0 (0.0%)
1010	18 (1.3%)	5 (2.2%)	13 (1.2%)	9 (3.9%)	4 (0.7%)	17 (1.1%)	5 (2.1%)	17 (1.2%)	3 (1.3%)	3 (0.4%)	20 (7.8%)
1011	36 (2.5%)	9 (3.9%)	33 (3.0%)	9 (3.9%)	8 (1.3%)	32 (2.0%)	3 (1.2%)	19 (1.3%)	2 (0.8%)	11 (1.3%)	6 (2.4%)
1100	44 (3.1%)	3 (1.3%)	51 (4.6%)	9 (3.9%)	41 (6.9%)	43 (2.8%)	10 (4.1%)	82 (5.6%)	9 (3.8%)	46 (5.4%)	2 (0.8%)
1101	36 (2.5%)	7 (3.0%)	27 (2.5%)	3 (1.3%)	9 (1.5%)	38 (2.4%)	4 (1.7%)	23 (1.6%)	1 (0.4%)	11 (1.3%)	1 (0.4%)
1110	51 (3.5%)	9 (3.9%)	15 (1.4%)	10 (4.3%)	7 (1.2%)	65 (4.2%)	4 (1.7%)	34 (2.3%)	6 (2.5%)	10 (1.2%)	4 (1.6%)
1111	242 (16.8%)	36 (15.5%)	202 (18.3%)	38 (16.3%)	121 (20.4%)	193 (12.4%)	25 (10.3%)	210 (14.3%)	32 (13.4%)	138 (16.1%)	16 (6.3%)
n	1439	232	1102	233	593	1562	242	1473	239	855	255

Table 4: All 16 four-branch outcome patterns per cell: count (share of the cell's Bank P states). Pattern digits are $(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$. KV4r: Q/K-rotated KV4; W3 and KV4r cells are stress cells with 3 blocks.

$P(I \neq 0)$. The control re-pairs states independently within a cell while keeping both marginals, $p_e = \sum_k p_Q(k)\, p_B(k)$, and Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$. Pools use the paper's equal weight per cell over common complete blocks, with the same bootstrap draws as Table 3. This is a pairing agreement control, not a no-interaction null and not a chance baseline for controller dependence (§4). In the W4 pool the control expects 38.0% [36.7, 39.0] off-diagonal against the observed 26.9%, so $\kappa = 0.29$ [0.27, 0.32] (new-task [0.22, 0.36]); in the KV8 pool it expects 33.6% ($\kappa = 0.55$). The effects under the two controllers agree more than random cross-state pairing would. On BFCL and ALFWorld the observed shares also lie below the control (expected 16.2–32.4% and 42.9–46.4% in the W4 cells; κ 0.47–0.71 and 0.14–0.24; Table 7), and on the telecom holdout the pooled 20.7% compares with 25.7% expected ($\kappa = 0.19$; Table 18).

F.2 Mean squared interaction under the second simulator (exploratory)

On the 80 states of the Bank P simulator control (Appendix D.4), each arm has $n_a = 5$ replicates under Sonnet 5.5. Per state, $\hat{I}_s = \bar{Y}_{BQ} + \bar{Y}_{QB} - \bar{Y}_{BB} - \bar{Y}_{QQ}$ and $U_s = \hat{I}_s^2 - \sum_a \bar{Y}_a(1-\bar{Y}_a)/(n_a - 1)$, which is unbiased for I_s^2 if, given the state, the replicates of each arm are i.i.d. Bernoulli and the four arms are independent (no shared cache or seed). Pooled with equal weight over the four W4 cells, mean $\hat{I}_s^2$ is 0.110, the noise correction 0.060 and mean U_s 0.050, with 95% intervals [0.004, 0.107] over states and [0.001, 0.111] over task clusters. One cell is negative (q4 AWQ-W4, −0.025) and is kept, as are negative per-state values. The estimate is the mean squared probability-scale interaction over these 80 frozen states. It is not the share of states with interaction and not a variance, it is not comparable with the deterministic 26.9% of §4, and we make no significance

Cell	G	H	F	$G-H$	$F-G$	$F-H$
q4 GPTQ-W4	$[-10.3, +4.2]$	$[-4.0, +6.0]$	$[-9.9, +4.7]$	$[-11.2, +3.6]$	$[-4.7, +6.1]$	$[-11.1, +4.4]$
q4 AWQ-W4	$[-6.5, +6.1]$	$[-3.7, +5.5]$	$[-7.4, +5.8]$	$[-7.7, +5.1]$	$[-6.2, +5.4]$	$[-8.7, +4.7]$
q8 GPTQ-W4	$[-3.3, +10.5]$	$[-3.8, +4.9]$	$[-2.0, +13.0]$	$[-3.2, +9.3]$	$[-3.0, +7.1]$	$[-2.0, +12.5]$
q8 AWQ-W4	$[-4.1, +8.7]$	$[-7.1, +4.6]$	$[-6.2, +7.3]$	$[-1.6, +8.6]$	$[-7.6, +4.2]$	$[-3.8, +7.5]$
W4 pool	$[-2.9, +3.8]$	$[-2.7, +2.8]$	$[-3.1, +4.4]$	$[-3.1, +3.7]$	$[-2.8, +3.1]$	$[-3.5, +4.5]$
q4 KV8	$[-9.4, +7.3]$	$[-6.4, +1.1]$	$[-9.3, +7.7]$	$[-7.1, +9.7]$	$[-5.7, +5.3]$	$[-6.7, +9.5]$
q8 KV8	$[-4.0, +6.9]$	$[-3.5, +2.9]$	$[-3.5, +8.1]$	$[-3.8, +7.2]$	$[-2.5, +5.0]$	$[-3.1, +8.7]$
KV8 pool	$[-5.3, +5.6]$	$[-3.7, +0.9]$	$[-4.5, +5.8]$	$[-3.8, +6.6]$	$[-2.8, +3.9]$	$[-3.0, +7.3]$
q4 GPTQ-W3^s	$[-9.7, +6.6]$	$[-6.6, +14.5]$	$[-6.8, +9.4]$	$[-15.0, +4.0]$	$[-5.2, +10.5]$	$[-12.2, +6.6]$
q4 KV4-rot^s	$[-3.4, +14.2]$	$[+2.1, +17.0]$	$[+3.0, +21.6]$	$[-13.5, +7.0]$	$[-1.3, +17.1]$	$[-6.2, +12.2]$
q8 GPTQ-W3^s	$[-6.7, +12.1]$	$[-7.2, +8.1]$	$[-6.6, +11.8]$	$[-7.7, +11.2]$	$[-9.1, +9.1]$	$[-6.8, +11.0]$
q8 KV4-rot^s	$[-6.6, +13.8]$	$[-11.2, +8.5]$	$[-6.6, +15.9]$	$[-3.3, +12.2]$	$[-7.2, +9.9]$	$[-2.5, +14.6]$
q8 RTN-W3^s	$[-4.7, +3.9]$	$[+5.9, +25.9]$	$[+9.4, +28.2]$	$[-24.7, -7.1]$	$[+10.2, +27.8]$	$[-7.1, +13.3]$

Table 5: New-task generalisation: crossed task×block bootstrap 95% intervals for the estimates of Table 3 (pp; same Bank P states). These answer whether the effect would hold for new tasks drawn like the panel; the fixed-pool intervals of Table 3 treat the 85 tasks as fixed. ^{s}stress cell, 3 blocks.

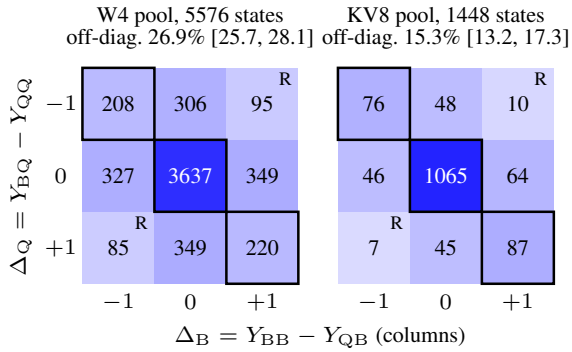


Figure 8: Does the action swap have the same effect under both controllers? Counts of (Δ_Q, Δ_B) over the Bank P states of the W4 cells (left) and the KV8 cells (right). Boxed: diagonal. R: strict reversals. Shading: share of states (power scale). Headers give the off-diagonal share as the equal-weight average over cells, the aggregation used throughout the paper except for the state-weighted menu oracle of Appendix J, with fixed-pool 95% intervals.

Cell	states	off-diag. %	exp. %	κ [fixed pool]	κ [new tasks]
q4 GPTQ-W4	1102	25.0 [22.8, 27.3]	40.5	0.381 [0.325, 0.437]	[0.245, 0.520]
q4 AWQ-W4	1439	29.3 [27.2, 31.2]	38.5	0.238 [0.215, 0.261]	[0.138, 0.343]
q8 GPTQ-W4	1473	22.9 [21.1, 24.5]	36.3	0.369 [0.324, 0.409]	[0.235, 0.492]
q8 AWQ-W4	1562	30.5 [28.3, 32.6]	36.8	0.171 [0.128, 0.213]	[0.075, 0.274]
W4 pool	5576	26.9 [25.7, 28.1]	38.0	0.291 [0.265, 0.318]	[0.223, 0.359]
q4 KV8	593	15.7 [12.7, 18.6]	36.2	0.567 [0.487, 0.643]	[0.412, 0.705]
q8 KV8	855	14.9 [12.9, 17.0]	31.1	0.522 [0.462, 0.584]	[0.386, 0.656]
KV8 pool	1448	15.3 [13.2, 17.3]	33.6	0.546 [0.487, 0.604]	[0.436, 0.652]
q4 GPTQ-W3^s	232	31.5 [10.5, 52.6]	36.1	0.129 [−0.258, 0.534]	[−0.049, 0.347]
q4 KV4-rot^s	233	33.0 [17.6, 48.3]	38.0	0.131 [−0.016, 0.280]	[−0.034, 0.298]
q8 GPTQ-W3^s	242	23.1 [6.0, 40.2]	36.4	0.364 [0.190, 0.548]	[0.199, 0.532]
q8 KV4-rot^s	239	31.0 [19.0, 43.0]	37.7	0.179 [0.098, 0.251]	[0.032, 0.327]
q8 RTN-W3^s	255	29.0 [24.6, 33.5]	28.6	−0.016 [−0.233, 0.200]	[−0.101, 0.142]

Table 6: Pairing agreement control per cell (Bank P divergent states). Off-diag.: observed share with $\Delta_Q \neq \Delta_B$ (fixed-pool 95%); exp.: share expected under marginal-preserving within-cell re-pairing. Pools: equal weight per cell. ^{s}Stress cells (3 blocks; t intervals; flagged, never pooled). In q8 RTN-W3, $\Delta_Q = 0$ on 93.3% of states, so p_e is close to p_o and $\kappa \approx 0$.

claim.

F.3 Replacement and four-branch estimands on BFCL and ALFWorld (supplementary)

Table 7 gives the replacement quantities of §4 (r, d, M and G_{all} over all items, non-divergent items counting as zero) and applies the four-branch analysis to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within benchmark category). The four-branch columns are post hoc and not part of any pre-registered analysis. In the KV4 cells $Y_{QQ} = 0$ on every state, so Δ_Q is almost constant, their off-diagonal mass is H and κ is near 0 by construction. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%); pairing κ is in Appendix F.1. The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ $[-43.5, -25.4]$ pp; KV4 cells reduce to H.

G Coverage Matrix

Table 8 lists the completed crossed-intervention coverage behind every result. It counts tasks, states and complete blocks after the strict admission rule.

H Bank U per Sub-Bank

Table 9 gives the per-state turnover behind Appendix D.4, over blocks f01–f03. Exact action agreement is 0.43–0.83 per cell and sub-bank.

Text content. Over f01–f03, an above-median relative length difference between the two texts

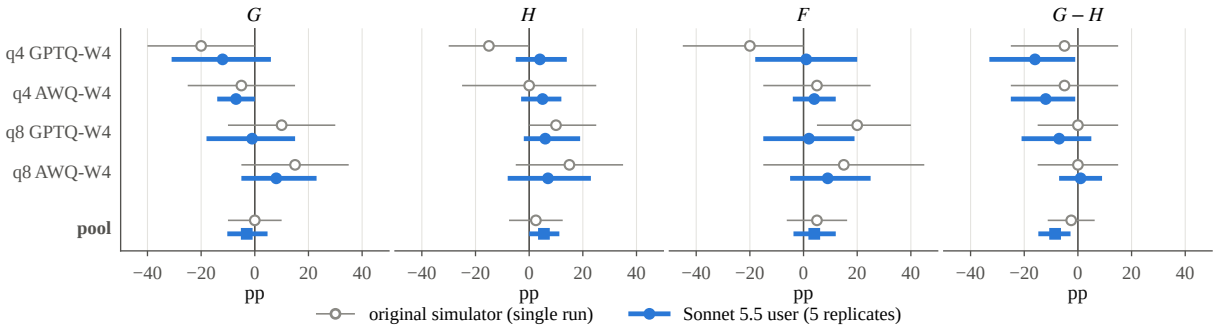


Figure 9: Bank P simulator control: G, H, F and $G - H$ on the 80 sampled Bank P states (20 per W4 cell), under the original greedy simulator (single runs; open circles) and with Sonnet 5.5 as the user (five replicates per arm; filled). pp; 95% intervals over states; pool: equal weight per cell. The sample is too small to test few-pp contrasts (Appendix D.4).

	all items					divergent items							
Cell	items	r	d	M	G_{all}	states	G	H	F	$G - H$	off-diag. %	exp. %	κ	rev. (n)
BFCL q4 GPTQ-W4	788	6.3	4.3	10.7	+2.0 [−0.3, +4.3]	573	+2.8 [−0.4, +5.9]	+1.4 [−0.7, +3.5]	+4.2 [+0.9, +7.2]	+1.4 [−1.8, +4.5]	11.7	27.4	0.574	3
BFCL q4 AWQ-W4	788	5.7	9.5	15.2	−3.8 [−6.5, −1.3]	642	−4.7 [−7.8, −1.5]	−1.1 [−3.6, +1.4]	−1.4 [−4.8, +1.7]	−3.6 [−6.7, −0.2]	17.1	32.4	0.471	8
BFCL q8 GPTQ-W4	788	2.9	2.3	5.2	+0.6 [−0.9, +2.3]	533	+0.9 [−1.3, +3.3]	+1.5 [+0.0, +3.2]	+2.1 [−0.4, +4.6]	−0.6 [−3.1, +2.0]	4.7	16.2	0.710	1
BFCL q8 AWQ-W4	788	4.1	2.7	6.7	+1.4 [−0.3, +3.3]	548	+2.0 [−0.4, +4.7]	+1.8 [+0.0, +3.7]	+3.6 [+0.9, +6.5]	+0.2 [−2.6, +3.1]	7.8	19.9	0.607	0
BFCL W4 pool					+0.1 [−1.0, +1.2]		+0.3 [−1.2, +1.8]	+0.9 [−0.2, +2.0]	+2.1 [+0.3, +3.9]	−0.6 [−2.3, +1.0]	10.3			
ALFWorld q8 GPTQ-W4	134	9.0	12.7	21.6	−3.7 [−11.2, +4.5]	115	−4.3 [−13.3, +5.0]	+7.0 [−2.6, +16.7]	+9.6 [−0.9, +19.7]	−11.3 [−20.9, −1.8]	40.0	46.4	0.137	4
ALFWorld q8 AWQ-W4	134	10.4	14.2	24.6	−3.7 [−11.9, +4.5]	119	−4.2 [−13.6, +4.9]	+5.0 [−2.6, +12.2]	+4.2 [−5.9, +13.7]	−9.2 [−18.8, +0.0]	32.8	42.9	0.237	3
ALFWorld W4 pool					−3.7 [−9.7, +1.9]		−4.3 [−11.1, +2.2]	+6.0 [−0.5, +12.6]	+6.9 [−2.1, +15.5]	−10.3 [−17.7, −2.9]	36.4			
BFCL q4 GPTQ-W3^s	591	6.1	3.6	9.6	+2.5 [+0.2, +5.1]	540	+2.8 [+0.2, +5.5]	+5.2 [+2.9, +7.7]	+8.9 [+5.7, +12.2]	−2.4 [−5.7, +0.7]	13.0	23.3	0.443	1
BFCL q8 GPTQ-W3^s	788	5.2	4.1	9.3	+1.1 [−1.0, +3.3]	696	+1.3 [−1.1, +3.7]	+1.4 [−0.1, +2.9]	+3.4 [+0.9, +6.0]	−0.1 [−2.6, +2.4]	8.3	20.4	0.592	1
BFCL q4 KV4^s	788	0.3	0.0	0.3	+0.3 [+0.0, +0.6]	787	+0.3 [+0.0, +0.6]	+2.9 [+1.9, +4.1]	+20.3 [+17.6, +23.0]	−2.7 [−3.9, −1.5]	20.5	20.9	0.020	0
BFCL q8 KV4^s	788	0.3	0.0	0.3	+0.3 [+0.0, +0.6]	788	+0.3 [+0.0, +0.6]	+1.8 [+1.0, +2.7]	+14.5 [+12.1, +17.0]	−1.5 [−2.5, −0.6]	14.7	15.2	0.029	0
ALFWorld q8 GPTQ-W3^s	134	6.7	3.0	9.7	+3.7 [−1.5, +9.7]	130	+3.8 [−1.5, +9.8]	+38.5 [+29.8, +46.9]	+42.3 [+33.6, +50.8]	−34.6 [−43.5, −25.4]	30.8	34.1	0.098	0
ALFWorld q8 KV4^s	134	0.0	0.0	0.0	+0.0 [+0.0, +0.0]	134	+0.0 [+0.0, +0.0]	+38.1 [+30.6, +45.5]	+53.0 [+44.8, +61.2]	−38.1 [−45.5, −30.6]	26.9	26.9	0.000	0

Table 7: Replacement and four-branch estimands on BFCL multi-turn and ALFWorld (no LLM user simulator; one run per item; 95% item-bootstrap intervals). All items: rescue r, damage d, turnover M (% of all items) and net G_{all} (pp), non-divergent items counting as zero. Divergent items (supplementary, exploratory, post hoc; pp, divergent items): Off-diag.: share with $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$; exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals. Pools: equal weight per cell. ^{s}Stress and unrotated KV4 cells; in KV4 $Y_{\text{QQ}} = 0$ on every state, so the table reduces to H.

co-occurs with outcome change in U1 and U2B of both models (flipping pairs among above- vs. below-median pairs: q4 U1 30/154 vs. 15/155; q8 U1 22/101 vs. 6/101; q4 U2B 33/138 vs. 11/139; q8 U2B 16/99 vs. 9/100). In f01 alone this was visible only for q4. Question, confirmation-request and new-entity flags show no consistent association. The features are regular expressions, and this is descriptive.

Bank U simulator control: sensitivity and scope. Dropping two mechanically flagged replies leaves the reading of Appendix D.4 unchanged (6, 6 and 25 of 37; Table 11). In either direction, 13 of the 39 original flip states flip again, against 11 of 39 new flips at matched non-flip states; this does not show that the two groups are equivalent. The relay's responders may have seen other requests. The control's scope is narrow: fixed histories from the original simulator, a stochastic relay rather than API Sonnet or real users, block f01 only and three replicates.

Shared fragility across quantizers (a lead). On the 255 shared q4 U2B states of f01–f03, cell pairs co-flip 21 times, against about 3.3 expected under independence, and 16 of the 21 co-flips go in the same direction. They come from 15 distinct states, 3 of which flip under all three cells. On q8 U2B, where every f01 flip was single-cell, 10 states now flip under two or more cells (14 pairwise co-flips against about 2.8 expected; 10 of 14 in the same direction). Joint flips almost never share an identical quantized action across cells (0 of 21 in q4, 2 of 14 in q8). We still read this as a lead toward *state-level* fragility, where different actions at the same state flip the outcome, and not yet as a finding: the co-flips are counted on one fixed set of states per block, with no correction.

A common controller (Q0). On the frozen 25% subsample of U2 in f01–f03, also continued by a common controller Q0, B's action leads to the identical outcome in every cell on all 146 shared states, which checks determinism. Turnover un-

Cell	τ^2-bench retail (Bank P)	τ^2-bench airline (Bank P)	telecom dev (Bank P)	Bank U (retail / airline)	BFCL / ALFWorld
q4 GPTQ-W4	58 t · 727 st · 20/20 b	27 t · 375 st · 20/20 b	holdout only	3 b · 393 / 177 rows	788 items · 573 st / –
q4 AWQ-W4	58 t · 986 st · 20/20 b	27 t · 453 st · 20/20 b	–	3 b · 393 / 177 rows	788 items · 642 st / –
q4 KV8	58 t · 412 st · 22/22 b	27 t · 181 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q8 GPTQ-W4	58 t · 1017 st · 20/20 b	27 t · 456 st · 20/20 b	80 t · 477 st · 6/6 b; holdout	3 b · 393 / 177 rows	788 · 533 st / 134 · 115 st
q8 AWQ-W4	58 t · 1087 st · 20/20 b	27 t · 475 st · 20/20 b	80 t · 480 st · 6/6 b; holdout	3 b · 393 / 177 rows	788 · 548 st / 134 · 119 st
q8 KV8	58 t · 585 st · 22/22 b	27 t · 270 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q4 GPTQ-W3	58 t · 162 st · 3/3 b	27 t · 70 st · 3/3 b	–	–	591 · 540 st† / –
q4 KV4-rot	58 t · 164 st · 3/3 b	27 t · 69 st · 3/3 b	–	–	– / –
q8 GPTQ-W3	58 t · 166 st · 3/3 b	27 t · 76 st · 3/3 b	–	–	788 · 696 st / 134 · 130 st
q8 KV4-rot	58 t · 165 st · 3/3 b	27 t · 74 st · 3/3 b	–	–	– / –
q8 RTN-W3	58 t · 174 st · 3/3 b	27 t · 81 st · 3/3 b	–	–	– / –
q4 / q8 KV4 (unrotated)	–	–	–	–	788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st
q14 GPTQ-W4	–	–	–	–	788 · 471 st / –

Table 8: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (Appendix H). "holdout": the cell is in the telecom holdout (Appendix L). Qwen3-32B has end-to-end runs only; Llama-3.1-8B stopped at its fork gate (Limitations); the q14 (Qwen3-14B) BFCL row is a within-family scale check. q4 RTN-W3 has no complete Bank P block and is omitted. The telecom development results (Appendix D.3) use the 6 blocks listed. †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix F.3). The end-to-end q4 AWQ-W3 run is incomplete (two airline episodes hit 600 s serving timeouts), and so are the q4 RTN-W2 stress runs (context-window overflows of the degenerate model and a truncated result file); neither is reported.

Cell	U1 (n=255)	U2B (n=255)	U2Q (n=60)
q4 GPTQ-W4	.090 (11/12)	.082 (10/11)	.083 (4/1)*
q4 AWQ-W4	.082 (7/14)	.078 (7/13)	.083 (3/2)
q4 KV8	.047 (4/8)	.039 (5/5)	.017 (0/1)
q8 GPTQ-W4	.047 (8/4)	.067 (9/8)	.050 (3/0)*
q8 AWQ-W4	.086 (14/8)	.090 (15/8)	.067 (3/1)
q8 KV8	.035 (5/4)	.031 (6/2)	0 (0/0)

Table 9: Bank U, blocks f01–f03: per-state turnover M (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). G lies within ± 0.028 in U1 and U2B and within ± 0.050 in U2Q, and every interval includes 0. Intervals are t intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]). *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals.

Model:sub	(i) identical	(ii) same tool structure, text	(iii) different tool or args	(iv) act vs. talk
q4:U1	399: 0	309: 45 (.146)	19: 1 (.053)	38: 10 (.263)
q8:U1	483: 0	202: 28 (.139)	47: 11 (.234)	33: 4 (.121)
q4:U2B	430: 0	277: 44 (.159)	24: 4 (.167)	34: 3 (.088)
q8:U2B	470: 0	199: 25 (.126)	40: 9 (.225)	56: 14 (.250)
q4:U2Q	108: 0	63: 10 (.159)*	5: 0	4: 1 (.250)
q8:U2Q	129: 0	33: 6 (.182)*	11: 1 (.091)	7: 0
unique rows	1943: 0	1045: 150 (.144)	143: 26 (.182)	169: 32 (.189)

Table 10: Which kind of action difference changes the outcome at ordinary states (Bank U, blocks f01–f03). Entries: "state rows: outcome-changing rows (flip rate)", pooled over the three cells of a model within a sub-bank. The last row counts each (cell, state) once: q4/q8 GPTQ-W4's U2Q states repeat their U1 states (* includes 5 and 3 repeated class-(ii) flips), so 216 flipping rows become 208. In block f01 alone the corresponding counts were 39 of 56 flips in class (ii), with rates .126/.109/.200 for (ii)/(iii)/(iv); the 58 f01 flips scored in Appendix D.5 include the two duplicated U2Q rows.

der Q0 is not consistently above or below the native turnover on the same rows (q4 AWQ-W4 U2B 8/58 vs. 5/58; q8 AWQ-W4 U2B 4/52 vs. 5/52; q4 KV8 U2B 1/58 vs. 2/58), so the f01 observation that Q0 turnover is at least the native one does not hold over three blocks. With 17–58 states per cell and sub-bank, this is too small to say whether the controller changes which actions matter.

I Concentration of Turnover in Tasks

This appendix completes the exploratory concentration result of §4 (development data, 85 tasks × 20 blocks). The per-task dispersion of M is 4.0–8.7 times its binomial null in the W4 and KV8 cells, and the top 20% of tasks hold 57–71% of all flips. We split the 20 blocks into two disjoint halves of 10 nonces. Task-level M correlates across halves (Spearman 0.52–0.86). Tasks that were net-rescue in one half have a mean net gain in the other half that is 14–51 pp higher than that of net-damage tasks, in all 12 cell-directions. A Freedman–Lane test conditions on measured baselines: QQ success propensity and BF16 success, plus trajectory length in a post-hoc variant. With them, one half's task rate still predicts the other's in all 6 configurations × 2 directions for both M and G (Holm-adjusted). This is a *retrospective conditional association* on viewed development data. It is carried by retail. Airline alone gives

Analysis	flips	retained	rev.	vanished (s/f/tie)	ctrl	new flips (resc./dam.)
All replies	39	7 [0.07, 0.34]	6	26 (15/2/9)	39	11 [0.15, 0.45] (7/4)
Flagged dropped	37	6 [0.06, 0.32]	6	25 (15/2/8)	39	11 [0.15, 0.45] (7/4)

Table 11: Bank U simulator control (block f01; Sonnet 5.5 relay user, $R = 3$). Fates use the sign of the replicate-mean flip indicator; s/f: both arms succeed/fail. "Flagged dropped": the two arm replicates with a mechanically flagged reply are set to missing. Replicates of one arm disagree in 16/39 (BQ) and 16/39 (QQ) flip states and in 10/39 and 8/39 control states. Per single replicate, retained flips are 5, 7 and 4 of 39.

8 of 12 for M (5 after Holm), and 3 of 12 under a heteroscedasticity-robust wild bootstrap. Direction persistence beyond flip opportunity holds more often for rescues than for damages, in 11 versus 6 of the 12 cell-directions. Different quantizers flip the same tasks: the task-identity null is rejected in 7 of 8 quantizer pairs. When both flip, however, they agree in direction only 42–81% of the time. This overlaps with the stable failure set of Jang et al. (2026); unlike that work, we record the direction of each flip.

J Trigger Headroom and a KL Reference Row

Table 12 compares the frozen trigger of §6 (frozen rule, unchanged) with two references on the same 1020 f02/f03 U1 rows, which hold 38 rescues and 31 damages. The *matched action-replacement oracle* selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits. The *three-option menu oracle* $E[\max(G, H, F, 0)]$ is the share of states where some arm beats QQ. On Bank P divergent states it is 0.202 over the four W4 cells (state-weighted, the one exception to the paper's equal-weight pools) against 0.117 for the action alone (1127 vs. 654 of 5576 states), and 0.191 vs. 0.113 over the six core cells. It is hindsight selection that also allows different replacement types, with different costs, so it is not a same-cost deployment comparison and not comparable with the action rule.

The *KL reference row* is a post-hoc auxiliary baseline. Its score is a truncated $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ over the union of both models' top-20 next-token sets plus a tail bucket, per a_{Q} token, teacher-forced on identical prompt ids. It needs a BF16 prefill of the prompt and a_{Q} at every decision (here 6709.3 prompt and 82.3 action tokens on average), so it is not Q-only. It captures turnover above chance (33 vs. 16.7 at 20%) but no significant net gain (Holm $p = 1.0$). Three aggregations over the a_{Q} tokens were fit on f01 only, with per-cell thresholds and the frozen rule's selection criterion; the maximum was chosen, and all of f02/f03 is reported. The two candidates not chosen on f01 look better on f02/f03 (e.g. the mean at 20%: net $+9$, Holm $p = .0852$), but selecting them now would be selection on validation data. Because the f02/f03 outcomes had been seen, the row is post hoc, and it is not a DMR reproduction, which trains a router.

K Controlled Injection and Process Audit

A natural explanation of the two-sided turnover is a local-defect story: the quantized agent makes a local error, and the task fails unless the continuation recovers. Two analyses constrain this explanation without identifying a mechanism.

Controlled defect injection. We froze 30 ordinary states from q4 GPTQ-W4 trajectories, all from one development block. At each state we forced one of three actions as the next step: an INValid action, a matched VALid action, or a process-perturbation control (PP) that repeats the last successful read. Either B or Q then continued. The defect categories were wrong_id, wrong_tool and premature_write, with 10 states each. Every state passed a replay audit of 22 recorded fields (VAL×Q against the original episode). Three fields were not recorded: the counters right after the restore, the rendered-prompt hash sequence and the tool-schema hash sequence.

Result: two-sided turnover, no clear error-specific difference detected. Injected defects change the outcome in 7/30 states under B (4 rescues, 3 damages) and 4/30 under Q (2 and 2). The process-perturbation control changes 5/30 under both controllers (Table 13). The error-specific effect, INV minus PP, is 0.00 [−0.13, +0.13] under B and −0.03 [−0.20, +0.10] under Q. The controller dependence D is nonzero in 8 of 30 states, but its mean is +0.03 [−0.17, +0.27]. So injected defects cause heterogeneous two-sided turnover, and at $n = 30$ states we detect no clear difference from a generic process perturbation and no mean controller difference. The intervals are wide, so

Budget	same per-cell selection counts sel.	oracle r/d/net (pp)	frozen r/d/net (pp)	captured	KL reference (max. per-token KL; needs BF16; post hoc) sel.	net vs. random (p; Holm)	r/d	turnover vs. random (p)
5%	50	34/0/+34 (+3.33)	6/2/+4 (+0.39)	4/34	72	−1 vs. +0.45 (.8337; 1.0000)	5/6	11 vs. 4.58 (.0038)
10%	114	38/0/+38 (+3.73)	7/5/+2 (+0.20)	2/38	126	+1 vs. +0.82 (.5483; 1.0000)	8/7	15 vs. 7.54 (.0048)
20%	222	38/0/+38 (+3.73)	16/11/+5 (+0.49)	5/38	257	+3 vs. +1.59 (.3984; 1.0000)	18/15	33 vs. 16.73 (.0000)

Table 12: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts) and a post-hoc KL reference row, on f02/f03 U1 (1020 rows). Captured: frozen net over oracle net. Random: matched-count per-cell random-trigger expectation; p: exact one-sided hypergeometric, Holm over the three budgets.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	4/3	7/30	+0.03 [−0.13, +0.20]
B	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.20]
B	INV vs PP	2/2	4/30	0.00 [−0.13, +0.13]
Q	INV vs VAL	2/2	4/30	0.00 [−0.13, +0.13]
Q	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
Q	INV vs PP	2/3	5/30	−0.03 [−0.20, +0.10]
Controller dependence D ($\neq 0$ in 8/30 states)				+0.03 [−0.17, +0.27]

Table 13: Defect injection on 30 frozen q4 states (development, one block): per-state rescue/damage of one arm against another, under the BF16 (B) or quantized (Q) continuation controller. $D = [Y(\text{INV}, B) - Y(\text{VAL}, B)] - [Y(\text{INV}, Q) - Y(\text{VAL}, Q)]$. Task-clustered bootstrap intervals; all exact McNemar $p \geq 0.25$.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
B	PP vs VAL	4/2	6/30	+0.07 [−0.07, +0.23]
B	INV vs PP	1/2	3/30	−0.03 [−0.13, +0.07]
Q	INV vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
Q	PP vs VAL	5/0	5/30	+0.17 [+0.03, +0.30]
Q	INV vs PP	2/6	8/30	−0.13 [−0.30, +0.03]
Controller dependence D ($\neq 0$ in 6/30 states)				0.00 [−0.17, +0.17]

Table 14: Defect injection on 30 frozen q8 GPTQ-W4 states (development, one block), as in Table 13. Task-clustered bootstrap intervals; exact McNemar $p \geq 0.0625$.

this does not show that errors and generic perturbations are equivalent.

Manipulation checks limit the categories. Two of 30 states failed a manipulation check: an injected write was rejected, and a valid reference itself returned "not found". Restricting to the 28 passing states leaves the reading unchanged. The wrong_tool category mostly produced argument errors: 8 of 10 injected calls failed with an unexpected-keyword error. We therefore call it a *tool-call/schema error perturbation*, not a wrong-tool test. premature_write is mostly failure→failure: the valid arm under Q already fails in 7/10 of these states, and turnover is 0/10 under both controllers. The event chain error → correction → related requirement satisfied completes in only 6 (B) and 7 (Q) of 29 injected errors, although 17 are corrected under each controller. Injections add tool errors (+0.43 under B, +0.57 under Q per branch).

Replication on q8 GPTQ-W4. The same design on 30 frozen q8 states (one development block; 10 per category) gives the same reading (Table 14). All 30 states pass the replay audit. Injected defects change the outcome in 5/30 states under each controller (3 rescues, 2 damages), the process-perturbation control in 6/30 (B) and 5/30 (Q). The error-specific effect is −0.03 [−0.13, +0.07] under B and −0.13 [−0.30, +0.03] under Q; the latter interval still allows a sizeable error-specific harm, and D is nonzero in 6 of 30 states with mean 0.00 [−0.17, +0.17]. Under Q, the process-perturbation control itself rescues 5 states and damages none (+0.17 [+0.03, +0.30], McNemar $p = 0.0625$). Manipulation checks fail more often than in q4: 7 of the 10 premature writes failed the check (6 were rejected by the environment, 1 executed without changing the database), and 2 valid references returned "not found". On the 21 passing states, INV vs VAL turnover is 5 (B) and 4 (Q), and $D \neq 0$ in 5. The q8 premature_write category therefore mostly did not inject a write.

Process audit (LLM-annotated). We sampled 80 development fork states, 20 per outcome sign. Two independent Sonnet 5.5 instances annotated them blind, a third instance adjudicated, and rule-based programmatic labels were added. These annotations are a *process description* conditional on outcome-defined groups. They are not causal evidence, not prevalences and not human labels. Three patterns stand out (Table 15). First, local defects and final failure do not correspond one-to-one. In damage states, where Q's action led to success and B's to failure, Q's first-action defect is corrected later in 10 of 20 QQ branches. Second, whether a write executes

does not separate the failing from the succeeding continuation of the same state (rescue 17 vs. 17; damage 15 vs. 14). Third, the failing continuation more often carries the rubric label `unintended_or_unconsented_change` (rescue: QQ 9 vs. BQ 2, paired difference $+0.35$ $[+0.08, +0.57]$; damage: 1 vs. 9, -0.40 $[-0.62, -0.11]$). In damage states, 7 of these 20 BQ branches carry the label on writes *with* recorded consent. That is the semantic "wrong change" part of the label, and it overlaps with the database-defined outcome by construction. Loops and budget exhaustion also mark failing branches (rescue: QQ 5 vs. BQ 0). We make no claim about consent. The programmatic consent field agrees poorly with the annotators, so we use it only as a mechanical row. Re-labelling 30 states with fresh instances of the same model family reproduces the key pattern.

K.1 Process audit table

The core "local defect" combines groundedness, argument validity and permission/consent judgements. The tool-validity field (inter-annotator $\kappa = 0.29$) is excluded from it. Mechanical fields agree with the programmatic verifier (write executed vs. the rubric's irreversible-change field, $\kappa = 0.99$). The consent fields agree only moderately ($\kappa = 0.44$ on writes both see).

L Holdout Pre-Registration

The protocol was frozen before any holdout unit was run, and its hash was recorded in the split manifest. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- two fixed halves of six blocks each;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside ± 3 pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical. Table 16 gives the confirmatory result. Table 17 gives all six stratum families (Holm within family), which are descriptive. Every level estimate is ≤ 0, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues (-9.09 pp) and the fewest-faults tercile (-6.37 pp).

Post-hoc four-branch estimands on the holdout. Table 18 is a supplementary, descriptive output computed after the confirmatory analysis, with the same admission and units. It is not part of the pre-registered confirmation and changes neither the primary endpoint nor its reading. Its G is a ratio over divergent states, whereas G_{all} in Table 16 averages over all admitted units with non-divergent units counted as 0.

M Withdrawn, Negative and Supplementary Results

W×KV interaction (not supported). A first q4 analysis found a W4×KV4 interaction of -14.9 $[-23.8, -6.0]$ pp ($k = 3$). Its KV16 reference lacked the Q/K rotation used for KV4, so rotation and quantization were confounded. With a matched rotated reference, the q4 interaction is -9.0 $[-30.6, +12.5]$. The pre-registered q8 replication gives $+2.1$ $[-5.8, +10.1]$ over 6 blocks, crossed $[-8.2, +12.4]$. The replication is not supported, and we make no W×KV claim.

Static precision restoration (appendix only). We tested whether restoring selected modules to higher precision recovers a damaged RTN-W3 model. We report this line as a mechanism stress test, not a method. In RTN-W3 the gains largely tracked the number of restored bits. A random MLP arm also recovered much of the gap, and no arm beat whole-model GPTQ-W4 at matched storage. A q8 RTN-W3 comparison of restored attention against byte-matched random attention is incomplete, and we report no result from it. The W4 restoration comparison below is a separate, pre-specified design.

Success equivalence. Not established (Table 1).

Supplementary perturbation analyses (exploratory). These analyses perturb the quantized weights directly instead of forking

Label (source)	Comparison	rescue	damage	both succeed	both fail
`unintended_or_unconsented_change` (LLM)	QQ vs BQ	9 vs 2, $+0.35$ [$+0.08$, $+0.57$]	1 vs 9, -0.40 [-0.62, -0.11]	1 vs 1	5 vs 7, -0.10 [-0.35, $+0.17$]
write executed (programmatic)	QQ vs BQ	17 vs 17, 0.00 [-0.25, $+0.25$]	15 vs 14, $+0.05$ [-0.26, $+0.35$]	10 vs 9	11 vs 13
executed write w/o recorded consent (prog.)	QQ vs BQ	9 vs 5, $+0.20$ [-0.06, $+0.43$]	3 vs 3, 0.00 [-0.21, $+0.21$]	3 vs 2	4 vs 3
loop or budget exhaustion (LLM)	QQ vs BQ	5 vs 0, $+0.25$ [$+0.04$, $+0.47$]	0 vs 2, -0.10 [-0.30, $+0.08$]	1 vs 2	7 vs 4
action judged indefensible (LLM)	Q vs B action	7 vs 8, -0.05 [-0.37, $+0.28$]	12 vs 5, $+0.35$ [-0.05, $+0.65$]	10 vs 6	12 vs 9
local defect, core (LLM)	Q vs B action	4 vs 6, -0.10 [-0.33, $+0.14$]	12 vs 7, $+0.25$ [-0.09, $+0.53$]	6 vs 3	11 vs 7
QQ first-action defect corrected later (LLM)	single branch	4/20 [0.08, 0.42]	10/20 [0.30, 0.70]	4/20	7/20

Table 15: Process audit of 80 development fork states, 20 per outcome sign (rescue: $Y_{\mathrm{BQ}} = 1, Y_{\mathrm{QQ}} = 0$; damage: $Y_{\mathrm{BQ}} = 0, Y_{\mathrm{QQ}} = 1$). **LLM-annotated** by Sonnet 5.5 (two independent blind instances and an adjudicating instance), plus rule-based programmatic labels; no human labels. Counts are first vs. second of $n = 20$ states. Intervals are within-state paired differences with Newcombe 95% CIs; single-branch rows use Wilson 95% intervals. This is a process description conditional on outcome-defined groups: not causal, not prevalence. The label name is the rubric category; we make no claim about consent.

Cell (weight)	units	excl.	G	r	d	ρ_c
q4 GPTQ-W4 (1/2)	396	1	-3.03 [-7.86, $+1.80$]	$+5.05$	$+8.08$	0.668
q8 GPTQ-W4 (1/4)	408	0	-5.64 [-9.33, -1.94]	$+4.41$	$+10.05$	0.374
q8 AWQ-W4 (1/4)	396	1	-6.57 [-11.98, -1.15]	$+6.31$	$+12.88$	0.644
Weighted (G_{all})			-4.57 [-7.17, -1.96]	$+5.21$	$+9.77$	0.588

Table 16: Telecom holdout, pre-registered analysis (34 tasks $\times$ 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional (G_{all}, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search).

Family	Level	n	G [crossed 90%]	p	Holm p
step budget[a]	T1	138/129/110	-0.32 [-1.57, $+0.94$]	.6771	.6771
	T2	124/126/162	-2.30 [-3.88, -0.72]	.0165	.0495
	T3	123/144/124	-1.94 [-3.35, -0.54]	.0229	.0495
persona	Hard	11	-5.99 [-12.23, $+0.26$]	.1147	.3355
	None	12	-3.65 [-7.42, $+0.13$]	.1118	.3355
	Easy	11	-4.36 [-9.19, $+0.48$]	.1385	.3355
BF16 difficulty	0	20	-2.35 [-4.09, -0.62]	.0257	.0772
	low	11	-4.54 [-8.90, -0.19]	.0860	.1720
	high	3	-18.75 [-38.00, $+0.50$]	.1091	.1720
task length	T1	9	-5.09 [-9.81, -0.38]	.0755	.1511
	T2	8	-0.78 [-4.46, $+2.90$]	.7270	.7270
	T3	17	-6.17 [-10.49, -1.84]	.0189	.0567
issue type	mms	14	-1.15 [-3.33, $+1.04$]	.3880	.4434
	mobile data	11	-9.09 [-14.98, -3.20]	.0111	.0333
	service	9	-4.40 [-10.32, $+1.52$]	.2217	.4434
faults	T1	19	-6.37 [-9.91, -2.83]	.0030	.0091
	T2	10	-1.94 [-7.58, $+3.69$]	.5704	.5704
	T3	5	-2.92 [-5.18, -0.66]	.0337	.0674

Table 17: Telecom holdout strata: the six pre-registered families (descriptive; Holm within each family over its three levels). pp. For task-level families, n is the number of tasks and G is G_{all} over the tasks in the level. [a]Step budget (remaining steps at the divergence) is unit-level: n is the number of divergent units per cell (q4 GPTQ-W4/q8 GPTQ-W4/q8 AWQ-W4), and G is the level's contribution to G_{all}, so the three levels sum to G_{all}. Tertile cut points were fixed on telecom development tasks.

trajectories, and they enter no claim in the main text. Each derived checkpoint takes every decoder linear from the BF16 model, from Q, or from an interpolation between them; all other tensors come from the BF16 model, and the endpoints reproduce BF16 and Q bit for bit. *Restoration location versus random modules.* On the GPTQ-W4 and AWQ-W4 checkpoints, restoring the BF16 weights of one layer third, of attention only or of the MLP only is compared with restoring randomly chosen modules with the same number of BF16 parameter bytes. [result pending: closed-loop success of located vs. byte-matched random restoration, with intervals]. *Error dose.* The residual intervention $W_\alpha = W + \alpha(W_{\mathrm{Q}} - W)$ scales the quantizer's own weight error, with $\alpha \in \{0, 0.25, 0.5, 1, 2\}$; it is not a bit width. [result pending: action agreement with BF16 and closed-loop success as functions of α]. *Low-bit failure modes.* [result pending: how W3 and lower-bit checkpoints fail in closed loop, by failure category].

N Provenance

Every number in the paper is traceable to released analysis outputs. Figures are drawn by a released plotting script from those outputs, with assertions on every value the text quotes, and the TikZ figures are typeset from numbers in the LaTeX source.

Cell (weight)	states	G	H	F	$G - H$	mean I	off-diag. %	exp. %	κ	rev. % (n)
q4 GPTQ-W4 (1/2)	385	$-3.1\,[-9.0, +2.8]$	$+0.5\,[-5.4, +6.4]$	$+1.8\,[-4.8, +8.4]$	$-3.6\,[-9.1, +1.8]$	$-4.4\,[-12.5, +3.7]$	19.7	27.0	0.269	3.4 (13)
q8 GPTQ-W4 (1/4)	399	$-5.8\,[-10.2, -1.3]$	$-4.0\,[-10.2, +2.2]$	$-5.3\,[-11.6, +1.1]$	$-1.8\,[-6.4, +2.9]$	$-4.5\,[-10.4, +1.4]$	19.5	23.1	0.155	3.0 (12)
q8 AWQ-W4 (1/4)	396	$-6.6\,[-13.0, -0.1]$	$-8.3\,[-15.3, -1.3]$	$-8.3\,[-15.5, -1.2]$	$+1.8\,[-3.3, +6.8]$	$-6.6\,[-14.5, +1.4]$	23.7	26.0	0.088	1.0 (4)
Weighted pool	1180	$-4.6\,[-7.8, -1.5]$	$-2.8\,[-6.3, +0.6]$	$-2.5\,[-6.2, +1.3]$	$-1.8\,[-4.8, +1.1]$	$-5.0\,[-9.6, -0.4]$	20.7	25.7	0.195	2.7

Table 18: Telecom holdout, *post-hoc* four-branch estimands on divergent states (descriptive; not part of the pre-registered confirmation). pp; crossed task$\times$block 95% intervals; pool weights as pre-registered. Exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals.